

EVOLUTION OF SEROPOSITIVITY IN PATIENTS TREATED FOR SLEEPING SICKNESS: IMPLICATIONS FOR INTERRUPTION OF TRANSMISSION.

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I. ABBREVIATIONS

BBB	Blood Brain Barrier
CATT	Card Agglutination Test for Trypanosomiasis
CNS	Central Nervous System
CSF	CerebroSpinal Fluid
CTC	Capillary Tube Centrifugation
ELISA	Enzyme-Linked ImmunoSorbent Assays
g-HAT	Gambiense Human African Trypanosomiasis
HAT	Human African Trypanosomiasis
IFU	Instructions For Use
LAMP	Loop-mediated isothermal AMPlification
mAECT	mini Anion Exchange Centrifugation Technique
NECT	Nifurtimox-Eflornithine Combination Therapy
NSSEP	National Sleeping Sickness Elimination Program
PCR	Polymerase Chain Reaction
PNLTHA	Programme National de Lutte contre la Trypanosomiasis Humane Africaine
RDT	Rapid Diagnostic Test
r-HAT	Rhodesiense Human African Trypanosomiasis
TL	TrypanoLysis test
VAT	Variable Antigen Type
VSG	Variant Surface Glycoprotein
WHO	World Health Organization

II. ABSTRACT

Introduction. Human African Trypanosomiasis (HAT), or sleeping sickness, is a neglected infectious disease caused by two subspecies of *Trypanosoma brucei*. HAT consists of two differentiated stages: at first, the parasite is infecting the blood and lymph, until it eventually reaches the brain, causing strong neurological alterations. Its diagnosis relies first on the identification of serological patients, followed by parasitological confirmation and staging. Gambiense HAT (g-HAT) is the most widespread subspecies, and its control strategies mainly rely on the diagnosis and treatment of its human reservoir. In the last thirty years, a strong, multidisciplinary effort has brought its incidence to an all-time low; however, with decreasing incidence, better tests are needed to detect the few, remaining cases. In the last years, new serological tests have been developed. The objective of this study, set within the goal of interruption of transmission of g-HAT for 2030 set by the WHO, is to assess how the newly developed serological tests detect the serological response of treated patients, up to two years after treatment.

Materials and methods. ~~The following~~Five serological tests were performed in samples belonging to the THARSAT collection, that followed a cohort of 360 treated g-HAT patients for two years: HAT Sero-K-SeT, SD BIOLINE HAT, ELISA/*T. b. gambiense*, inhibition ELISA and immune trypanolysis test. The percentage of positivity was studied before and after treatment, and in five follow-up visits afterwards: 3, 6, 12, 18 and 24 months later.

Results. All the tests registered a higher percentage of negatives in the later follow-up visits (12, 18 and 24 months after treatment), but only SD BIOLINE HAT and ELISA/*T. b. gambiense* ~~detected—identified~~ over 50% of the samples as negative after two years. ~~Furthermore, even~~On the other hand, the SD BIOLINE HAT tests whose outcome was positive, ~~registered very faint lines~~ in the later follow-up visits, the intensity of the positive lines devlined and were mostly registered as very faint lines. HAT Sero-K-SeT consistently detected as positive almost all the samples, including those taken two years after treatment. Comparing all the tests by their probability of becoming negative in time, using trypanolysis as reference, showed that inhibition ELISA and trypanolysis had a similar performance, that HAT Sero-K-SeT had the highest probability of staying positive, and SD BIOLINE HAT and ELISA/*T. b. gambiense* the lowest.

Conclusion. Treated patients will remain positive at least two years after treatment, so it is important to rethink the diagnostic algorithm for places that have been highly endemic in the past. The combination of HAT Sero-K-SeT and SD BIOLINE HAT could help in telling apart treated and new infections. Inhibition ELISA showed a performance comparable to immune trypanolysis requiring less skill and resources. In future efforts for disease surveillance, the rapid tests are fit to be incorporated in passive screening in sentinel posts, and inhibition ELISA can be incorporated in bigger facilities for those cases where the diagnosis is unclear.

III. INTRODUCTION

Human African trypanosomiasis (HAT), also known as sleeping sickness, is a neglected tropical infectious disease caused by the parasite *Trypanosoma brucei*. The two subspecies that are able to infect humans are *T. brucei gambiense* and *rhodesiense*: the first one causes the slow-progressing, chronic form of the disease (called gambiense-HAT, or g-HAT), while the later leads to an acute, more fulminant form of sleeping sickness (known as rhodesiense-HAT, or r-HAT). Endemic in sub-Saharan Africa, HAT distribution follows that of its vector, the tsetse fly (genus *Glossina*); g-HAT is endemic in Central and Western Africa, while r-HAT is found in Eastern Africa [1].

HAT is a disease with a zoonotic origin: trypanosomes naturally infect animals, in which they may cause a disease known as nagana, or animal African trypanosomiasis. The parasite evolved, and eventually became able to also infect humans [2]. This shift probably happened thousands of years ago: the first reports of a disease similar to nagana were written by the Egyptians of the Old Kingdom, in papyrus dating from the 2nd millennium BC. However, it is not until the XII century that trypanosomiasis is described in humans, in travel journals of Arabs in close trading relationships with Western African kingdoms [3]. By the XVIII century, sleeping sickness was a widely known disease, closely related to slavery, although still little was known about what was causing it, or how to treat it.

In the early XX century, severe HAT epidemics shook the African continent, worsened by the ecological alterations and forced population movements brought by colonialism [1]. To limit its socioeconomic impact, colonial authorities set programs to control the disease: the discovery of suramin, used as treatment against acute HAT, and extended screening of the population succeeded in controlling the disease, bringing it to the brink of elimination in the 1960s [4]. The independence of most African countries brought an era of political and economic instability, with a disastrous effect on health services, particularly in control and prevention programs [5]. This, along with reduced disease awareness, led to a resurgence of cases that peaked in the 1990s, which brought sleeping sickness back to the public eye.

Strengthened control and surveillance in the past 20 years have again managed to progressively reduce disease transmission, reaching the milestone of less than a thousand new cases reported in 2018 [6]. National Sleeping Sickness Elimination Programs (NSSEPs), with the assistance of the World Health Organization (WHO), public-private partnerships with pharmaceutical companies and a range of other partners, have played an essential role in the control of sleeping sickness in the early XXI century. However, despite the drastic reduction in the number of new cases, HAT is still endemic in over a dozen African countries (14 countries reported cases of both gambiense and rhodesiense in 2018 [6]), causing significant morbidity in the affected rural populations [4].

a. EPIDEMIOLOGY

The transmission of HAT has lowered enough in the last twenty years that it is now considered a focal disease: its distribution is limited to a certain area, called focus, beyond which the disease does not occur [4]. From an epidemiological point of view, there is a clear

difference in the distribution of the two variants of the disease, as shown in **Figure 1**, that approximately matches the Rift Valley: g-HAT is found in the West, and r-HAT in the East.

Trypanosoma brucei gambiense is the most widespread subspecies, affecting several countries and causing over 98% of all the reported cases in 2018 [6]. In the last 9 years (2009 to 2018), the Democratic Republic of the Congo (DRC) has reported the majority of g-HAT cases, followed by Central African Republic, Chad, South Sudan and Angola. Other countries that are still endemic, although with a lesser incidence, are Cameroon, Congo, Côte d'Ivoire, Equatorial Guinea, Gabon, Guinea and Uganda. Burkina Faso, Ghana and Nigeria, where g-HAT was previously endemic, did not report any cases in 2018 [6]. Outside endemic areas, g-HAT cases are rare and associated to migrants and expatriates from endemic areas [7]. Although considered an anthroponotic parasite, it is true that gambiense infections can also affect domestic animals [8]. However, their role in g-HAT transmission remains unclear.

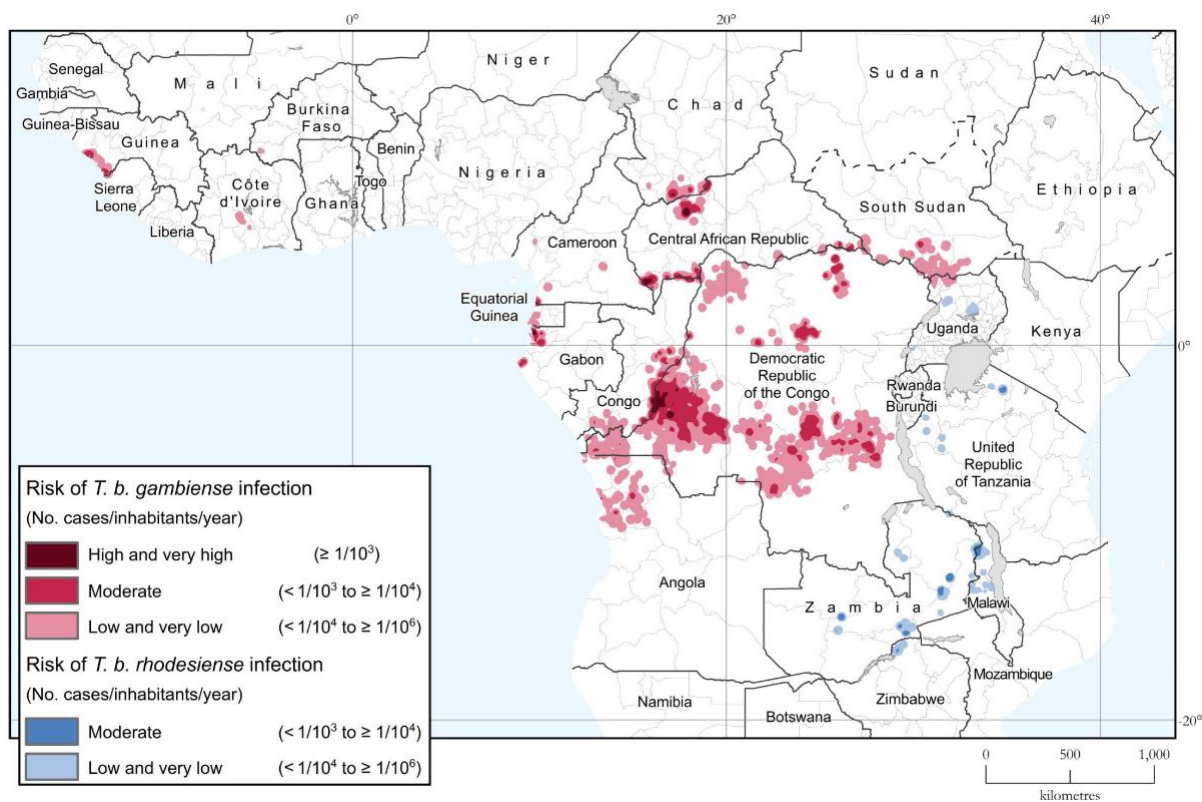


Figure 1. Geographical distribution of areas at risk of either *T. b. gambiense* or *T. b. rhodesiense* infection, from 2014 to 2018. A total of 54 million people are estimated to be at various levels of risk of infection by either sub-species of *Trypanosoma brucei*. Source: Franco et al (2020).

On the other hand, rhodesiense sleeping sickness is responsible for only a minority of the reported cases (2% in 2018 [6]). This subspecies naturally affects wild animals, and most human infections are related to sporadic, spill-over events from the animal reservoir [4]. Malawi, Uganda and Zambia still suffer of r-HAT, while other historically-endemic countries, like Kenya, United Republic of Tanzania and Zimbabwe, did not report any cases in 2018 [6]. It is worth noting that Uganda is the only country where both forms of the disease are endemic, although they do not overlap in any regions [9]. Outside of endemic areas, r-HAT cases are

more numerous than g-HAT, since they mostly affect tourists that go on safari to the aforementioned endemic areas.

Due to the differences in terms of epidemiology and biology between both forms of HAT, radically different strategies are needed for their control. For rhodesiense HAT, its elimination, understood as the total interruption of transmission, is considered impossible due to its zoonotic origin [10], and was not included in this study. For gambiense HAT, on the other hand, a huge progress has been made towards this end, with control strategies that mainly rely on the diagnosis and treatment of its human reservoir, as will be developed further on.

b. CLINICAL SYMPTOMS

Sleeping sickness develops in two distinct, separate stages. When the tsetse fly bites a person, it inoculates the trypanosomes, that multiply in the site of infection and then spread to the nearest lymph nodes [2]. At first, the parasite stays in blood and lymph, giving place to the haemolymphatic, or first, stage of the disease. The initial symptoms the patient manifests are intermittent fever, headaches and arthralgia; they are vague and non-specific, and can sometimes be mistaken with malaria, since both parasites have overlapping endemic areas [11]. As the parasite load grows, it disseminates through the body: the infection becomes systemic and new symptoms appear, like lymphadenopathy (which in g-HAT usually affects the posterior cervical region, and is known as Winterbottom's sign), pruritus, hepatosplenomegaly, oedema and endocrine dysfunction. Hormonal alterations have been described to cause amenorrhea, infertility, and even lead to miscarriage in women [1]. It is not the parasite itself causing the damage, but rather the immune response the host displays against it when trying to contain the infection [12].

Eventually, the trypanosomes cross the blood brain barrier (BBB) and invade the central nervous system (CNS), leading to the meningoencephalitic, or second stage of sleeping sickness [2]. It is characterized by neuropsychiatric disorders, out of which sleep disturbances are the most distinctive: there is a progressive disruption of the sleep-wake cycle, with daytime somnolence and uncontrollable urges to sleep, and nocturnal insomnia. Besides, the patient also suffers of motor and sensory alterations, as well as other, less-specific symptoms like irritability, fatigue and psychiatric and behavioural disturbances [13]. All the aforementioned neuropsychiatric disorders increase in severity and frequency as the disease progresses, leading to coma and, eventually, death [1].

Although the host immune response can have a heavy influence on the clinical evolution of HAT [14], the subspecies of *Trypanosoma brucei* causing the infection will ultimately determine how fast the patient moves from phase I to phase II. As mentioned before, *T. b. gambiense* causes a chronic infection: it takes, on average, three years for g-HAT to reach the brain and develop into the meningoencephalitic form of the disease, but it is subject to a great interpersonal variability [15]. Furthermore, although it is traditionally considered that, if left untreated, HAT is deadly, and that the progression to the meningoencephalitic stage is unstoppable, exceptions have been reported in some g-HAT patients, who not only survive without treatment, but are also able to somehow control the parasitaemia and remain asymptomatic for years [12].

It is important to determine if these patients, who remain seropositive but control the parasitaemia to levels below the threshold detectable by microscopy or polymerase chain reaction (PCR) [16], could constitute a human reservoir of gambiense sleeping sickness. If these asymptomatic patients can indeed transmit the disease, it would have important implications in the maintenance of transmission of HAT in a community, since these individuals would not be confirmed as cases in parasitological testing, and thus no treatment would be provided to them, leaving a door open for a resurgence of the disease after a time. To this day, the role of asymptomatic HAT patients in disease transmission remains unknown.

c. LABORATORY TESTING FOR g-HAT

The development of diagnostic tools for gambiense HAT has always been hindered by the extremely low parasite load that characterizes this form of the disease [17]. Nowadays, several diagnostic tests are available, that detect either the host's immunity against the parasite, the parasite itself, or its DNA or RNA. They are called, respectively, serological, parasitological, or molecular tests.

i. SEROLOGICAL TESTS

Trypanosoma brucei is an extracellular parasite: during an infection, it is continuously in contact with the immune response of the host. As a defence mechanism, trypanosomes have developed a dense coat of glycoprotein dimers, to protect themselves against attacks from the innate immune response. The coat is, however, highly immunogenic, and will elicit a strong and specific antibody response [13]. To avoid being eliminated by it, trypanosomes change their covers: a minority of the parasite's population express an antigenically different Variant Surface Glycoprotein (VSG). The antibody response will be directed against the previous variable antigen type (VAT), that is, the previous VSG, but will not be able to recognize the new one, so the parasite ensures the survival of at least part of its population, that will proliferate again [18]. Several VATs have been studied to be applied in diagnostics, as markers of the presence of the parasite. As of right now, the most widely used antigens are named LiTat 1.3 and LiTat 1.5: they are both VSGs that cover the external surface of the parasite in the bloodstream of the vertebrate host [19].

Serological tests use antigens of *Trypanosoma brucei gambiense* for the detection of antibodies produced against the parasite [20]. For the last decades, the most widely used serological test has been Card Agglutination Test for Trypanosomiasis (CATT). It consists of Commassie-blue-stained, LiTat 1.3-expressing *T. b. gambiense* cells as antigen, that is applied on a plastic agglutination card [18]. If the addition of a drop of blood reveals blue agglutination, that person has antibodies against g-HAT. CATT sensitivity on undiluted blood has reported to range between 68,8% to 100%, while its specificity goes from 83,5 to 99,3% [21]–[24]. CATT is simple, fast and cheap, and it has been crucial in the control programs deployed in the last decades, since it is designed for the mass screening of the population [21]. However, it is not appropriate for its implementation in peripheral health centres, with just a few suspected cases every day [25]: there are 50 CATT tests per vial, and open vials have to either be stored at 4° C (rarely doable in HAT-endemic areas) or discarded after 24 hours [26].

The decreasing incidence of sleeping sickness creates the need for highly specific diagnostic tests, but it also means the budget available for its research shrinks. Diagnostic tools have to adapt to the new epidemiological situation, focusing more on surveillance in healthcare centres than on active screening. In this context, new rapid diagnostic tests (RDTs) have been developed; namely, SD BIOLINE HAT and HAT Sero-K-SeT. They are immuno-chromatographic tests that allow the qualitative detection of antibodies after an incubation of 15 to 20 minutes [27]. The results can be read after 15 to 20 minutes. SD BIOLINE HAT has a sensitivity of 89,2% and a specificity of 94,58% [28], while HAT Sero K-SeT has a sensitivity of 98,5% and a specificity of 98,6% [25]. The advantage of RDTs in general is that they do not need a cold chain for their conservation or trained professionals for their performance: they are both cheap and easy to use. However, just like CATT, the specificity of RDT's is not 100%, which will lead to many false positives and low positive predictive values in areas with low prevalence [18].

Another important downside of the available RDTs, and of all the serological tests in general, is that they use native antigens: their production depends on the massive infection of laboratory animals with trypanosome clones expressing LiTat 1.3 and LiTat 1.5, that are highly infectious for humans [29]. In order to avoid the risk of infection of the staff and the need for laboratory animals in production, new RDTs are being developed, exclusively using recombinant antigens. A prototype of SD HAT BIOLINE 2.0 uses the antigens Invariant Surface Glycoprotein 65-1 (expressed in *Escherichia coli*) and VSG LiTat 1.5 (produced using a Baculovirus expression system) [30]. Recombinant versions of HAT Sero-K-SeT are also being developed, although they are not yet publicly available.

The only serological test that is considered to be 100% specific detecting antibodies against *T. b. gambiense* is the immune trypanolysis test (TL). Cloned live trypanosomes, expressing either LiTat 1.3 or LiTat 1.5 [31], are incubated with the serum of a patient and the serum of a guinea pig, which is rich in complement. g-HAT-specific antibodies from the patient's serum will bind to the parasite's surface, and the cell will be destroyed by antibody-dependent complement lysis [18]. The reported sensitivity for this test is 97,2 to 100%, while the specificity, as previously mentioned, is 100%. However, TL can only be performed in well-equipped, reference laboratories with very skilled personnel, meaning it is practically impossible to implement it in field conditions [32].

Enzyme-linked immunosorbent assays, or ELISAs, are another, very popular format of serological diagnostic tests, that has also been adapted for the serodiagnosis of g-HAT. ELISA/*T.b. gambiense* test consists of microplates covered with g-HAT VSG antigens, to which blood or serum from the patient is added [33]. The addition of anti-human antibodies and a substrate will lead to a measurable, colorimetric reaction that will reveal the presence or lack thereof of antibodies against the parasite in the patient's serum. ELISA/ *T.b. gambiense* has a sensitivity of 82,8% and a specificity of 99,8% [22].

Another format of ELISA has recently been developed: *T. b. gambiense* inhibition ELISA (g-iELISA), with an improved sensitivity and specificity, is able to achieve a similar diagnostic accuracy as TL, but with fewer disadvantages. g-iELISA is based on the principle of monoclonal antibodies competing with g-HAT patients antibodies for binding on the surface

glycoproteins that are coated on the microplates: the binding of the monoclonal antibodies is inhibited when human antibodies are present in the blood or serum sample from the patient. This strategy achieves higher sensitivity and specificity than regular ELISAs (94,7% and 100%, respectively), values that are comparable even to TL tests [34]. Both ELISA and iELISA are high throughput and easy to perform, but have the disadvantage of being demanding in terms of equipment, supplies and staff skills.

ii. PARASITOLOGICAL TESTS

Parasitological tests rely on the direct observation of the parasite. Although their specificity is considered to be 100%, since the detection of a trypanosome in a sample will unequivocally mean infection, their sensitivity is far from optimal, and varies greatly from test to test. It often depends on the equipment available and on the experience of the staff performing the tests. The simplest parasitological test is direct trypanosome detection in blood samples, using a microscope at maximum contrast and 40x magnification. The sensitivity of this method can vary between 3,9 and 54,2% [21]. If the necessary reagents are available, Giemsa staining can be applied to thick blood film preparations and the sensitivity increases from 25,9 to 100% [21], but it heavily depends on the skills of the laboratory technician and the reagents available. The problem is that, especially for g-HAT infections, the parasite load is so low it is often necessary to make several reads of the same sample, to confirm it is negative [35]. It is therefore not recommended for routine parasitological confirmation of HAT.

In order to work around the low parasitic loads in the samples, several separation and concentration methods exist. For example, the Capillary Tube Centrifugation (CTC), that consists on the centrifugation of blood in an heparinized tube, leaving the trypanosomes in the white cell zone, between the plasma and the erythrocytes. It has a sensitivity between 44,3 and 93%, but needs electricity, special equipment and trained technicians [21]. The reference technique recommended by WHO, however, is the mini Anion Exchange Centrifugation Technique (mAECT): live trypanosomes are separated from blood cells with an anion chromatography, and then they are concentrated at the bottom of the tube via low-speed centrifugation so they can be observed with a microscope [35]. This technique has a sensitivity of 80,4%, although it can be further improved using the buffy coat [36], obtained from the centrifugation of up to 10 mL of blood instead of plain blood to 90,9% [37].

Parasites can also be detected in lymph or cerebrospinal fluid (CSF). When the lymph node in the back of the neck is swollen, lymph node aspirate are performed, and the fluid is examined at 40x magnification in the microscope. The trypanosomes can be observed without using any reagent or staining. Lastly, trypanosomes can be observed in the CSF, however, it is not very common, as lumbar puncture is only performed for staging the disease since it is a highly invasive procedure [1].

iii. MOLECULAR TESTS

In the last 20 years, different polymerase chain reaction (PCR) tests have been developed for the detection of g-HAT, and although promising, they are still not widely available. PCR tests can be applied to blood, lymph or CSF samples. PCR on blood samples showed a sensitivity of 88% and a specificity of 99,2% [38]. The main problem with PCR-based methods

is that they often require electricity, specific equipment and expensive reagents. Low-tech alternatives, like loop-mediated isothermal amplification (LAMP) are also available: LAMP is able to amplify DNA under a constant temperature, usually in the range of 60 to 65° C, meaning that in 35 minutes LAMP is able to detect g-HAT using a simple water bath, with a sensitivity of 75% and a specificity of 100% [21]. It can be a very valuable tool for less equipped laboratories in rural endemic regions [39], but more research is still needed for its validation.

Molecular diagnostic tests have been available for the diagnosis of g-HAT for years, but none of them are widely used, since their implementation in healthcare centres in endemic regions is logistically difficult [7] and, since most of them target single-copy genes, their analytical sensitivity is poor [1]. Although molecular tests are generally considered sensitive and specific, for g-HAT, their reproducibility is not well documented, since the result of these techniques heavily depends on sample collection and storage [37], and the influence of contaminations and other factors.

d. PATIENT CARE: DIAGNOSIS AND TREATMENT

In order to provide treatment in the first stage of the disease, and avoid as much as possible the neurological complications that appear later on, a swift and accurate diagnosis is essential. However, diagnosis of HAT has proved to be challenging: in the first place, the clinical symptoms in both stages are non-specific [40], and can be easily mixed up with other infections, like malaria or filariasis [41]. Second, in situations of low prevalence (less than 1%), or where transmission has stopped altogether, the limited specificity and sensitivity of the available tests lead to low positive predictive values, and it becomes more and more difficult to detect the few, remaining cases [18]. Lastly, HAT is endemic in rural areas, often remote and isolated, where the health centres are ill-equipped and under-staffed [21], limiting the tests that can be implemented to those that do not need to maintain a cold chain or trained personnel, for example.

For now, g-HAT diagnosis follows a multi-step, laboratory-based decision tree [7], represented in **Figure 2** and divided in three stages. First comes the screening of the at risk population with serological tests, to find the individuals that have antibodies against the parasite and, thus, have been exposed to it. Traditionally, this first step has been done with CATT, although now the newly-developed RDTs are being introduced for the same purpose. Cases where the diagnostic is unclear are sent to reference laboratories, to perform trypanolysis or ELISA, since they are more specific. If antibodies against *T. b. gambiense* are detected by any method, the second step is the confirmation of the infection, usually by direct observation of the parasite with a microscope in lymph node aspirates or blood, or with concentration techniques like CTC or mAECT on blood. The third and last step is staging: up until now, lumbar punctures used to be systematically performed for staging the disease once the parasite was detected in the blood or lymph, to assess if the parasite had arrived to the brain. After centrifugation of the CSF sample, if trypanosomes or more than 5 white blood cells/mm³ were observed, the patient was considered to be in the second stage of the disease. Staging has always been necessary, since drug treatment regime differed according to the stage the patient was in, and the drugs available for the second stage of the disease are very toxic [17].

For the last years, the default treatment for g-HAT infections in the first stage has been pentamidine, given intramuscularly once a day for seven days. It is an effective and safe drug, although it does have adverse effects that include hypoglycaemia, hypotension, nausea and vomiting [42]. For second-stage infections of g-HAT, melarsoprol (an arsenic derivate that causes severe adverse effects) has been abandoned in favour of the much safer option of nifurtimox and eflornithine combination therapy (NECT), that also has a lesser treatment failure rate [43]. Although it is an improvement, it still requires patient hospitalization (nifurtimox is given orally, three times a day for ten days, and eflornithine is given intravenously, twice a day for seven days) and has important adverse effects [44].

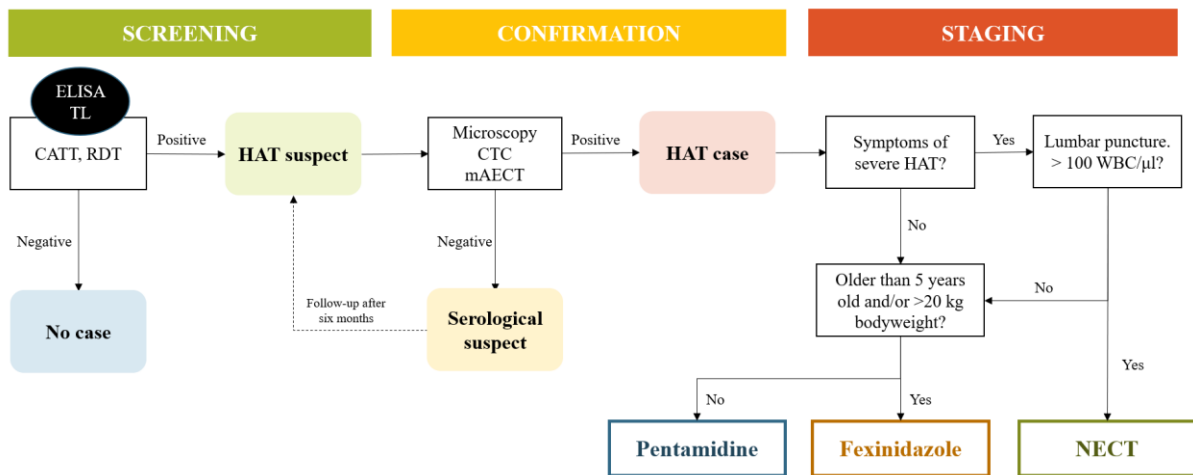


Figure 2. Latest diagnostic and treatment algorithm, following WHO guidelines determined in December, 2018 in Geneva, Switzerland [45].

However, in 2018, a new drug called fexinidazole, was approved as g-HAT treatment. It is administered orally, once a day for ten days, and it is effective against both stages of the disease [46]. Since the development of fexinidazole, the diagnosis and treatment decision tree has been revisited and adapted (**Figure 2**): now, fexinidazole is recommended for all patients in stage one, and patients without severe neurological involvement in stage two. Lumbar punctures are no longer systematically performed to every confirmed infection, but are relegated to those patients who have severe neurological symptoms of the disease. If trypanosomes are found in the CSF sample, or if the concentration of WBCs is greater than 100 per microlitre, NECT is recommended [47]. The one disadvantage of fexinidazole is that it has not been deemed safe yet for patients younger than six years old or that weigh less than 20 kilograms. In these cases, pentamidine is still recommended [45].

e. CONTROL EFFORTS

As mentioned before, in the last decades there has been a sustained decrease in HAT cases in the world, thanks to a renewed focus on the disease after a dramatic peak in reported cases in the 1990s [48]. Control strategies for g-HAT are based in mainly two aspects: first, the diagnosis and treatment of human cases, in order to reduce the parasite reservoir, and second, vector control, to reduce disease transmission [49]. With regards of the identification of g-HAT cases, two different strategies are followed, depending on the endemicity levels of the area:

passive screening, integrated in healthcare facilities, and active screening, carried out by mobile teams in areas of high incidence.

Active case finding is very effective in poor, remote HAT endemic areas that not always have access to healthcare facilities: mobile teams travel in groups of 7 to 9 people from village to village, performing mass serological screenings of the population using CATT. When a seropositive patient is identified, the procedure follows with parasitological confirmation (via lymph node aspirate, CTC or mAECT) and staging (via lumbar puncture). Depending on the stage of the disease the patient is in, the mobile teams will provide the necessary treatment or transport the patient to the nearest hospital. Although this strategy has been pivotal for disease control in the last 40 years [1], lately some changes have been introduced: for example, the so-called ‘mini mobile teams’, which are made up by less people and travel around in motorcycles. They typically consist of three screeners, performing serological tests, and one confirmer, that will visit the village where suspects were detected by the screeners. visiting endemic villages on motorcycles. Mini mobile teams normally use RDTs and mAECT for the diagnosis. [50]. When properly managed, three years of consecutive active screening can help greatly reduce transmission in a community.

As the prevalence of the disease decreases, mobile teams become more expensive: the cost per detected case increases, and it is more difficult to find funding [21]. In this new context of communities that are still endemic but with a very low prevalence, passive screening becomes of uttermost importance: g-HAT detection is integrated in the general health services in endemic areas, as a way to maintain a surveillance system in place [51]. RDTs could play a key role in this sense: with CATT, several samples need to be processed at once, while RDTs allow for the individual evaluation of different samples [52]. The main limitation of passive screening is that it mostly detects patients who are already in the meningoencephalitic stage [51], meaning that the symptoms are more serious, the patient faces a more toxic treatment and they contribute to disease transmission for a longer time. However, the detection of a confirmed HAT case through passive screening will trigger the screening of the village where the new patient resides, in order to detect any other possible cases and prevent the growth of a new infection focus [53].

In order to further reduce transmission and, hence, disease incidence, WHO recommends combining case detection strategies with vector control. Several tools are available for tsetse fly control in g-HAT endemic areas, mostly based on traps and insecticide impregnated targets. The environmental impact of these vector control strategies is considered acceptable, and besides, there is an economic benefit to be gained when reducing tsetse fly populations: not only human trypanosomiasis incidence decreases, but so does its animal variant, known as nagana, which has an important impact in cattle breeding communities [49].

In the last thirty years, the introduction and maintenance of these control strategies in g-HAT endemic areas have made a huge dent in the incidence and prevalence of the disease. The impact has been so great that, in 2012, WHO made the decision to target the elimination of HAT as a public health problem, meaning that less than 2,000 cases should be reported annually and a reduction by 90% of the new cases should be observed in at risk areas for 2020 [54]. In

2018, only 977 cases were reported of both gambiense and rhodesiense HAT, well below the objective set six years before.

Although for rhodesiense HAT elimination is currently considered impossible, due to its zoonotic nature and the limited availability of control tools [52], it is not the same scenario for g-HAT: in the new roadmap for Sustainable Development goals regarding neglected tropical diseases, WHO aims for the disruption of transmission of gambiense HAT (zero cases reported) by 2030 [55]. Besides the formerly mentioned control strategies, to achieve this goal, it is necessary to clear up certain uncertainties that still exist about gambiense sleeping sickness, such as the role of asymptomatic individuals in disease transmission. These are patients with a positive serological test but with very low levels of parasitaemia, that are often overlooked with low-sensitive methods like microscopy or CTC, so they appear to be parasitologically negative [14]. Not a lot is known about this status: what it can mean in terms of transmission, nor if they eventually develop HAT.

IV. OBJECTIVES

In the context of the 2030 goal of interruption of g-HAT transmission, it is important to understand how the available diagnostic tools will respond to the situations of decreasing prevalence. In a setting of a community that was previously endemic, where people have been infected and now passive screening-based surveillance systems are set, what can we expect of the available diagnostic techniques? Does a positive result in a previously-infected patient indicate a new infection, or just that the immunity is long-lasting?

The objective of this thesis was to assess the long-term serological status of treated g-HAT patients. Several serological tests, that are now used (RDTs) or are expected to become part of the serological screening process in the near future (ELISA, iELISA or TL) were performed in samples of over 300 patients, that were treated and followed up for two years. The goal of this retrospective study was to assess if treated patients eventually become serologically negative in the different available tests, or if they remain seropositive at least two years after treating the infection.

V. MATERIALS AND METHODS

The samples used for this thesis belong to the THARSAT study, which was carried out from May 2005 to May 2008 in the hospital of Dipumba, region of Mbuji Mayi (East Kasai Province, DRC [56]). This project followed a cohort of first- and second-stage g-HAT patients, from the moment of diagnosis to two years after treatment, in accordance to the WHO guidelines available back then. The samples were stored in liquid nitrogen and shipped to the Institute of Tropical Medicine in Antwerp (Belgium). Although they had been used for other studies, all the samples have been stored and maintained at -80 °C since.

a. PATIENTS

For recruitment to the THARSAT study, each patient underwent clinical examination and antibody testing with CATT. Seropositive individuals, or those with matching clinical symptoms, were tested for the presence of trypanosomes in lymph or blood, using CTC or mAECT. Staging was determined by counting leukocytes or proving the presence of parasites

after modified single centrifugation in the cerebrospinal fluid. Inclusion criteria included presence of trypanosomes in lymph, blood or CSF, being older than 12 years old and living within 100 kilometres from Mbuji-Mayi. After screening, 384 patients were diagnosed and eligible for the study, but 24 were ultimately excluded, based on the fact that they were pregnant, moribund, had haemorrhagic CSF, had another concurrent, serious disease or could not guarantee a follow-up.

Out of the 360 patients that were finally included in the study, 41 were in the first stage of the disease, and 319 in second stage. 242 were primary cases, meaning they had never been treated for HAT before, while 118 were secondary cases: they had been treated before, but relapsed. All the participants were treated, or retreated, following national guidelines (Programme National de Lutte contre la Trypanosomiase Humaine Africaine, PNLTHA): primary cases in the first stage of the disease were treated with pentamidine, whereas second-stage cases were treated melarsoprol or eflornithine. Retreatment cases received melarsoprol (monotherapy or in combination with nifurtimox) or eflornithine (monotherapy or in combination with melarsoprol or nifurtimox).

For monitoring, all the patients were asked to come back three, six, twelve, eighteen and twenty four months after treatment, and all the diagnostic tests were repeated in each visit, to assess treatment outcome. Thus, each patient should have seven samples: one before treatment (BT), one right after treatment (AT), and five in the follow-up. Since the goal of the present study is to assess the antibody response after treatment, those patients that had failed to complete at least one follow-up (that is, those patients that had less than three samples) were excluded from the analysis. 329 patients were considered eligible: 163 of them were deemed cured after the two year period, while 129 relapsed at some point during the follow-up (the rate of treatment failure with melarsoprol, which was administered to 55% of the patients, was unusually high in the region of Mbuji Mayi, with reports going up to 60% [56]). The other 39 patients either died during the follow-up (for causes related or not to HAT), voluntarily retreated from the study or failed to complete the follow-up.

Table 1. Distribution of samples: number of patients with samples available in each follow-up event.

	Before treatment	After treatment	3 months	6 months	12 months	18 months	24 months
All	329	321	312	266	193	150	166
Cured patients^a	163	159	156	156	145	132	159
Relapses^b	NA	NA	44	50	18	9	6

^a Number of samples of the 163 patients available for each follow-up. ^b Number of patients that relapsed in each date of the follow-up.

b. SEROLOGICAL TESTS

A total of 1,737 samples, belonging to 329 patients, were used for this work. The samples, that were stored at -80 °C, were defrosted by leaving them overnight at 4 °C. Then, they were centrifuged for five minutes, at 13.000 rpm and 4 °C. An aliquot of 120 µL was taken from

each sample onto a 96-well plate, and stored at -80 °C until further used. The following tests were performed for each sample.

RDTs. Two commercially available RDTs were used: HAT Sero-K-SeT, by CORIS BioConcept (Gembloux, Belgium), and SD BIOLINE HAT, by Abbot Standard Diagnostics (South Korea) (**Figure 3**). For HAT Sero-K-SeT, the sample application window is divided in two zones: the sample, 15 µL of plasma, was delivered in the upper zone, and then two drops of migration buffer (approximately 85 µL) are added in the lower zone. The test is incubated for 15 minutes inside the wrapper, to avoid the evaporation of the migration buffer. The test is invalid if the control line is absent, and negative if only the control line is present [57]. Additionally, two prototypes of recombinant RDTs from CORIS BioConcept, called respectively HAT Sero-K-SeT 2.0 Version 1 and Version 2, became available while performing the study. They were performed following the same instructions the manufacturer recommended for the native test, with the only difference that they could be incubated in open air (following the instructions of use (IFU) from the manufacturer). On the other hand, in SD BIOLINE HAT the sample application window comprises of one single zone: 10 µL of plasma sample was dispensed in the window, followed by four drops of chase buffer (approximately 120 µL). After incubating for 15 to 20 minutes, three lines may appear: the control line, that validates the test, and two test lines, one for the antigen LiTat1.3, and another for LiTat1.5. The presence of either test line means that the sample is positive [58].

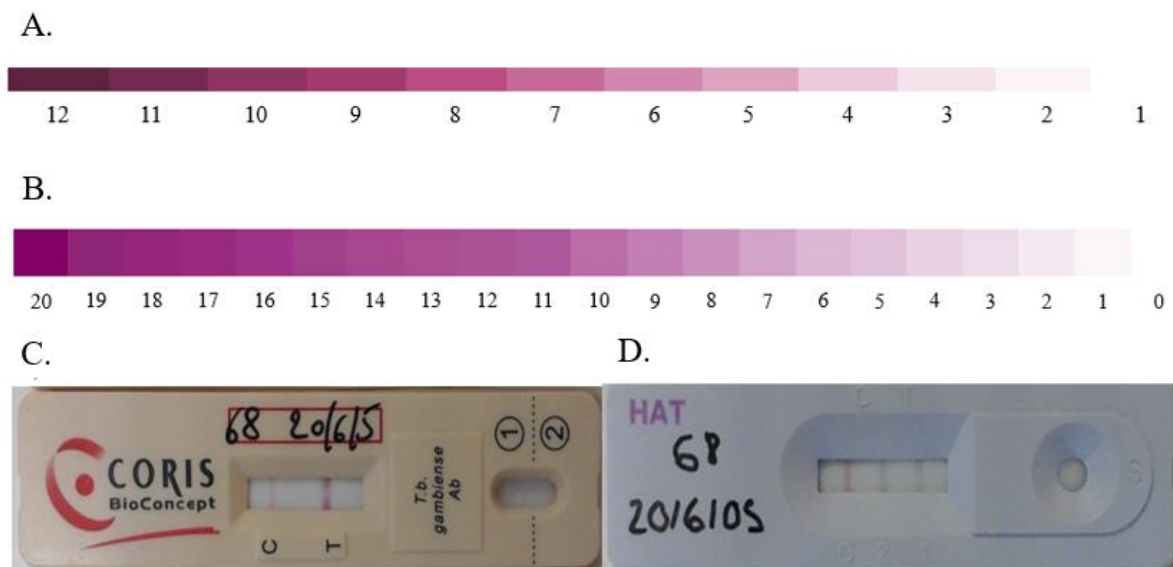


Figure 3. **A.** Representation of the intensity-score colour scale provided by the manufacturer of HAT Sero-K-SeT, (see picture in Appendix 1). **B.** Representation of the intensity-score colour scale provided by the manufacturer of SD BIOLINE HAT (see picture in Appendix 1). **C.** HAT Sero-K-SeT test, from left to right: Control line, Test line, Inoculation window divided in two spaces. **D.** SD BIOLINE test, from left to right: Control line, Test line 2 (corresponding to LiTat 1.5), Test line 1 (corresponding to LiTat 1.3), Sample well.

Both tests were read by two, independent readers (STA and CAS) that had no previous experience working with RDTs. Each reader recorded, for each test, the presence of a control line and the intensity of the test line (based on the colour scales in **Figure 3 A and B**): from 1 to 12, being 1 a negative result, in HAT Sero-K-SeT, and from 0 to 20 in SD BIOLINE HAT,

with 0 representing a negative result. Both RDTs were performed in five-sample rounds, and pictures were taken of both RDTs after incubation.

ELISA/T. b. gambiense. Half of the 96 well plates were coated with 150 μ L of a dilution of trypanosome antigens LiTat 1.3 and LiTat 1.5; each in a concentration of 2 μ g/mL in phosphate buffer (PB; 0.007M $\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$, 0.005M $\text{Na}_2\text{HPO}_4 \cdot 2(\text{H}_2\text{O})$, pH 7.4), giving a total protein concentration of 4 μ g/mL. The antigen-free wells were filled with 150 μ L of PB, and the plates were incubated overnight at 4 °C. Saturation of the plates was performed by adding 350 μ L of PBS-blotto (0.2M NaCl, 0.001M $\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$, 0.008M $\text{Na}_2\text{HPO}_4 \cdot 2(\text{H}_2\text{O})$, pH 7.15) to all the wells in the plate. After one hour incubation at room temperature, the wells were emptied and filled again with 350 μ L of PBS-sucrose (0.14M NaCl, 0.001M $\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$, 0.008M $\text{Na}_2\text{HPO}_4 \cdot 2(\text{H}_2\text{O})$, 5% sucrose, pH 7.15) for protecting the proteins before storing the plates at -80° C. When performing the tests, the plates were washed three times, with 350 μ L of PBS-Tween (0.14M NaCl, 0.001M $\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$, 0.008M $\text{Na}_2\text{HPO}_4 \cdot 2(\text{H}_2\text{O})$, 0.05% Tween detergent, pH 7.4). The plasma samples, along with the positive and negative controls were diluted 1 in 350 μ L in PBS-Blotto, and 150 μ L of each were added to both halves of the plate: the one coated with antigens and one antigen-free. After one hour incubation at room temperature, the plates were washed thrice with 350 μ L of PBS-Tween, and incubated with the conjugate (goat anti-human IgG (H+L) peroxidase/HRP conjugated (Jackson), diluted 1:40 000 (1:2 in glycerol, 1:20 000 in PBS Tween)) for 30 minutes at room temperature. Afterwards, the plates were washed five times with 350 μ L of PBS-Tween, 150 μ L of substrate (in this case, 3,3',5'5'-tetramethylbenzidine (TMB)) were added to each well, and incubated for 15 minutes. To stop the reaction, 150 μ L of sulfuric acid were added to each well. The photometric reading, at an absorption of 450 nm, was performed within 30 minutes after stopping the reaction [59], [60].

Corrected optical density (OD) values were obtained after subtracting the OD values of the antigen-free wells from the corresponding antigen-containing wells. For standardization, all results were expressed as percentage positivity (PP), proportionate to the positive control present in each plate. The PP was calculated as Percentage Positivity = (corrected OD of test sample/corrected OD of positive control) x 100 [60]. Quality control for each plate was performed using the positive and negative controls: the test was considered invalid if the corrected OD of the positive control was smaller than 1, if the corrected OD of the negative control was greater than 0.15 or if the PP of the negative control over the positive one was greater than 10%.

Inhibition ELISA. The commercial kit *Trypanosoma brucei gambiense* iELISA, developed by apDia (Turnhout, Belgium) was used to perform this test, according to the instructions provided by the manufacturer. As a summary, each test sample was first diluted 1:20: 17.5 μ L of sample in 332.5 μ L of the Sample dilution buffer provided with the kit. 150 μ L of each sample dilution was then added to both halves of the plate: one half coated with the antigen LiTat 1.3, and the other one with LiTat 1.5. 150 μ L of the negative and positive control were added in duplicates to both halves of the plate. The plates were incubated for one hour at room temperature, and then emptied and washed three times with 350 μ L of Wash solution. 150 μ L of Conjugate solution 1 (CONJ 1 LiTat 1.3) were added to the first half of the plate, and 150

µL of Conjugate solution 2 (CONJ 2 LiTat 1.5) to the remaining wells. Following a 30-minute incubation at room temperature, the plates were washed thrice with 350 µL of Wash solution, and 150 µL of the Chromogen solution were added per well. The plates were left to incubate for ten minutes, at room temperature but avoiding exposure to light. Then, after adding 150 µL of STOP solution, the absorbance of each well is to be determined at 450 nm, with reference filter 600-650 nm. [Supplementary material 5 of ref. 31].

To calculate the results, the OD-value of the Negative control was considered as having a 100% binding of the monoclonal antibodies. For each sample, the percentage of inhibition of monoclonal antibody binding (% inhibition) was calculated as follows: % inhibition = 100 – (OD sample / OD negative control x 100). A sample was considered positive if the % inhibition of either LiTat 1.3 or LiTat 1.5 was equal or greater to 30%, and negative if the % inhibition of both antigens was lower than 30%. In order to validate the results of each plate, the results of the controls were evaluated. The OD value of the negative control had to be between the values 1.000 and 3.100, and the % inhibition of the positive controls for both LiTat 1.3 and LiTat 1.5 should be greater than 30%. If one of the conditions was not met, the test was deemed invalid and had to be repeated [34].

Immune trypanolysis test. This test uses live, cloned trypanosomes that exclusively express either the antigen LiTat 1.3 or the antigen LiTat 1.5. For each sample, the test is performed in two rounds (one per antigen). In this study, the antigen LiTat 1.5 was performed for all the samples, since there was already existing data of LiTat 1.3 for the THARSAT collection. However, since that data was not fully completed, additional LiTat 1.3 trypanolysis tests were performed in the before treatment samples of those patients for whom trypanolysis had not been performed. Test preparations start with the infection of mice with the wanted clone (in this case, LiTat 1.5 clone), for it to replicate inside the animal. The clones of the parasite are kept frozen in liquid nitrogen: they need to be thawed in a bain-marie, at 37 °C, followed by the immediate addition of 1 mL of phosphate-buffer saline glucose (PSG, 0.03M NaCl, 0.002M NaH₂PO₄·H₂O, 0.04M Na₂HPO₄·2(H₂O), 10% anhydrous D-glucose, pH 7.9-8.1). Using phase-contrast microscopy, the concentration of the solution is estimated through matching method, and adjusted to the desired concentration for inoculation. Mice are infected with an intraperitoneal injection of 0.2 mL, and the parasitaemia is verified 48 to 72 hours after inoculation. The test can be performed when a parasitaemia of 8.1 to 8.4 via MM is achieved, but always within 96 hours after infecting the mice, in order to avoid antigen variation by the parasite in defence of the immune response of the mouse. For the test, 25 µL of guinea pig serum and 25 µL of each undiluted sample are added per well, along with 50 µL of a dilution of trypanosomes and guinea pig serum, adjusted with MM to 5 to 10 trypanosomes per field. A positive and a negative control need to be included in the plate too. The plate will be incubated for 60 minutes in a shaker, at room temperature. For the reading, 8 µL of the dilution of each well are pipetted on a microscope slide, and examined at x200 to estimate the percentage of trypanosomes alive per field. It is recommended to start with the controls: if in the negative control all the trypanosomes are alive, and in the positive control all the trypanosomes are dead, the test is validated. The trypanolysis test is considered positive when 50% of the trypanosomes, expressing either LiTat 1.3 or LiTat 1.5, are found dead. If over 50% of both LiTat 1.3 and LiTat 1.5 clones are alive, the sample is considered negative [31].

c. DATA ANALYSIS

Data was entered and organized using Microsoft Excel (Microsoft Corporation, 2018). The analysis and graph design was carried out using RStudio Version 1.4.1106 (Ed. R Foundation for Statistical Computing, Vienna, Austria). Quantitative variables were spread as median and interquartile range (IQR) if they did not follow a normal distribution, and as mean and standard deviation if they did. For the comparison of continuous numeric variables, the Mann-Whitney rank sum test was used, since none of them followed a normal distribution. For the comparison of qualitative variables, McNemar's test was used. The probability of a test remaining positive over time was calculated using the Kaplan-Meier survival function (as done in [61]), and the outcome was expressed as a frequency (from 0 to 1) with its 95% interval.

VI. RESULTS

a. RDTs

For each sample, both commercially-available RDTs were simultaneously performed on all the samples. However, due to the limited availability of the two versions of HAT Sero-K-SeT recombinant tests, these were only performed in the samples collected before treatment. The results are shown in **Table 2**: HAT Sero-K-SeT detected as negative 11 out of the 1,737 samples tested; six of them being samples taken two years after treatment. SD BIOLINE HAT, on the other hand, registered a higher number of negative results, following an increasing trend in time: while three months after treatment 89% of the sample were found positive, the percentage lowers to 66% after two years. Furthermore, the performance of each antigen is significantly different (p-value < 0.001): LiTat 1.3 reported a significantly higher amount of negative results in all follow-ups than LiTat 1.5, although this will be further explored later. For the recombinant versions of HAT Sero-K-SeT, there is an important improvement of Version 2 over Version 1: the former detected 99% of the before treatment (BT) samples as positive, while the later detected 84%.

Table 2. Number of negative results recorded in each rapid diagnostic test per follow-up visit.

		Before treatment (n=329)*	After treatment (n=321)	3 months (n=312)	6 months (n=266)	12 months (n=193)	18 months (n=150)	24 months (n=166)
Native	HAT Sero-K-Set	0 (100%)	0 (100%)	1 (100%)	1 (100%)	1 (99%)	2 (99%)	6 (96%)
	SD BIOLINE HAT	7 (98%)	12 (96%)	34 (89%)	46 (83%)	56 (71%)	54 (64%)	56 (66%)
	SD BIOLINE LiTat 1.3	90 (72%)	98 (69%)	159 (49%)	177 (33%)	150 (22%)	118 (21%)	136 (18%)
	SD BIOLINE LiTat 1.5	7 (98%)	13 (96%)	35 (89%)	49 (82%)	57 (70%)	59 (61%)	60 (64%)
Recombinant	HAT Sero-K-Set 2.0 Version 1	52 (84%)	-	-	-	-	-	-
	HAT Sero-K-Set 2.0 Version 2	4 (99%)	-	-	-	-	-	-

* n = number of patients in each follow-up event. In the table, number of negative results per test in n samples analysed, in parenthesis, percentage of positive samples in the same follow-up visit.

Since each reader wrote the intensity of the test lines per test, according to the colour scale shown in **Figure 3**, the evolution of the intensity of the test lines in the RDTs along the follow-up could be represented. As illustrated in **Figure 4**, HAT Sero-K-SeT showed a steady decrease in intensity over time: the median intensity score in samples before and after treatment was 11 (IQR: 10-11); however, as time went by, the median intensity significantly decreased. At three and six months, the median intensity score was 9 (IQR: 8-10, p-value<0.001 for 3 months, IQR: 7-10, p-value<0.001 for 6 months); and one year and one year and a half later, it further lowered to 8 (IQR: 6-9, p-value<0.001). Two years after the treatment, the median intensity score was 7 (IQR: 5-8, p-value<0.001).

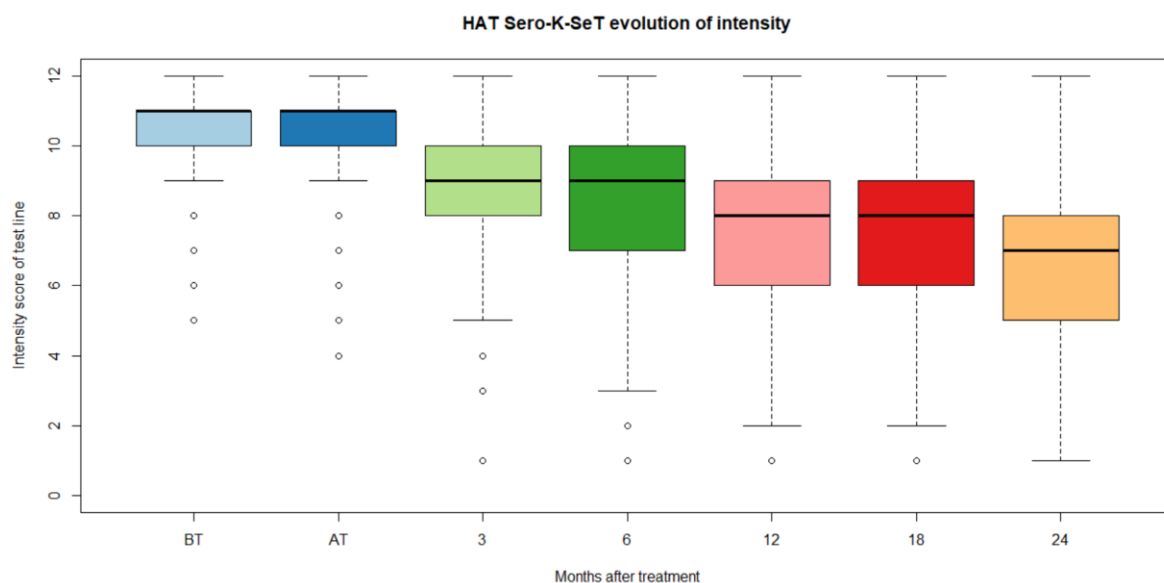


Figure 4. Evolution of the intensity score of the HAT Sero-K-SeT test over time. BT = Before treatment, AT = After treatment, 3 = 3-months follow-up, 6 = 6-months follow-up, 12 = 12-months follow-up, 18 = 18-months follow-up, 24 = 24-month follow-up.

The same procedure of writing down the intensity score of the test lines was followed for SD BIOLINE HAT, although the scale was different (from 0 to 20) and it had two separate test lines for the antigens LiTat 1.3 and LiTat 1.5. The evolution of the intensity score over the follow-ups was represented separately for each antigen in **Figure 5**. The antigen LiTat 1.3 registered very low intensity scores through all follow-ups: the median intensity scores in before and after treatment (AT) samples were 1 (IQR: 0-3, IQR: 0-2, respectively), and zero for the rest of the follow-ups (IQR: 0-1 for the first two, IQR: 0-0 for the rest). LiTat 1.5, on the other hand, had a wider variability and higher intensity scores, although still low if considered that the colour scale for this test was up to 20. The median intensity score BT was 5 (IQR: 2-7), and AT 4 (IQR: 2-6). From then on, it fell to 2 for the follow-ups at 3 and 6 months (IQR: 1-4 for both) and then it further lowered to 1 (IQR: 1-3 for 12 and 18 months, IQR: 1-2 for 24).

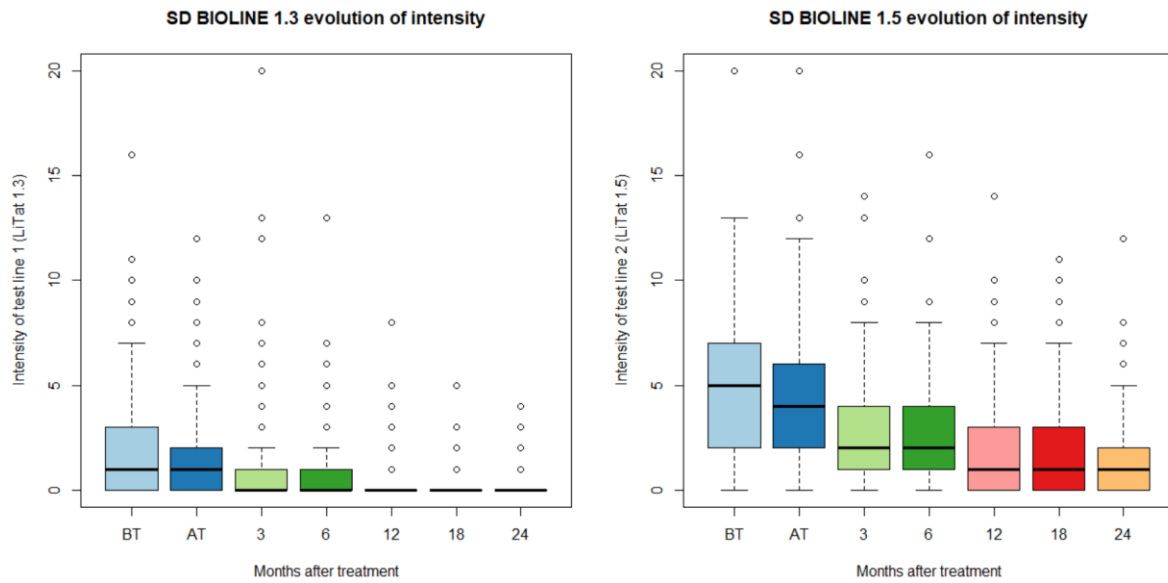


Figure 5. Intensity evolution in SD BIOLINE HAT test lines. On the left, evolution in the intensity of test line corresponding to the LiTat 1.3 antigen along the follow-up period. On the right, evolution in the intensity of the test line corresponding to LiTat 1.5 antigen along the follow-up period. BT = Before treatment, AT = After treatment, 3 = 3-months follow-up, 6 = 6-months follow-up, 12 = 12-months follow-up, 18 = 18-months follow-up, 24 = 24-month follow-up.

To further test the performance of both RDT models, the agreement between readers was evaluated through the observational analysis of disagreement described by Bland and Altman in [62]. For HAT Sero-K-SeT (**Figure 6**), the mean disagreement between readers was -0.41, with a standard deviation of 0.95. The discordance was highest in the higher values of the colour scale, which was considered normal since most tests reported values higher than 7. It was also noteworthy the agreement between tests in what constituted a negative result (in this case, a intensity score of one): no disagreement was reported in this value, although only 11 out of 1,737 samples were considered negative.

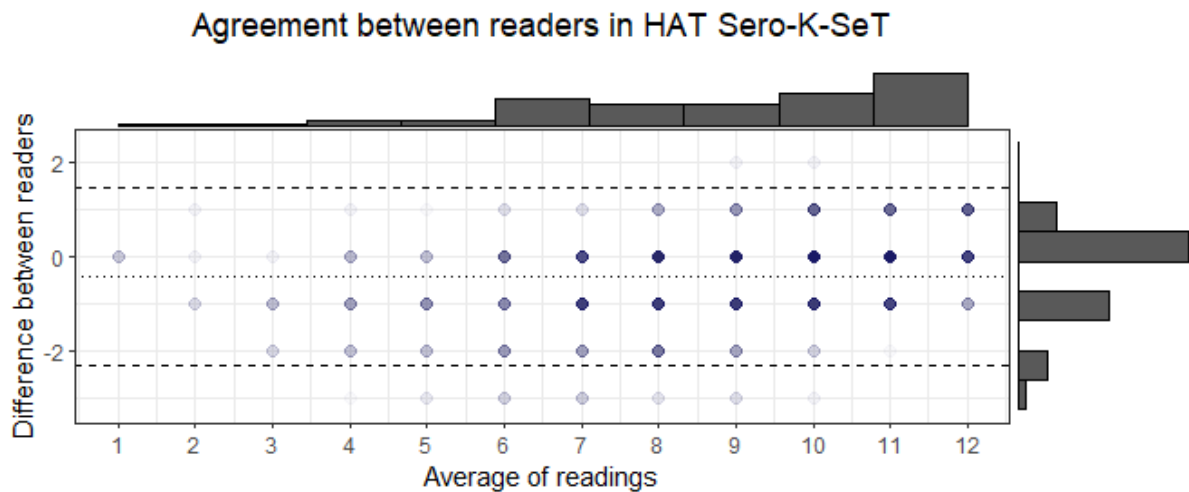


Figure 6. Agreement between readers in HAT Sero-K-SeT. The dotted line represents the mean difference, while the dashed lines represent the mean plus or minus two times the standard deviation. The histogram on top represents how common each value was in the tests, and the histogram on the right how common the discordance between readers was.

For SD BIOLINE HAT, the disagreement between readers was measured in two separate graphs, to represent each test line. For the first test line, corresponding to the antigen LiTat 1.3 and represented in **Figure 7**, the average difference between readers is -0.02, and the standard deviation is 0.67. The intensity score recorded most often in this test was zero, meaning that most SD BIOLINE HAT tests did not detect any antibodies against LiTat 1.3. Although most of the times the readers agreed on the intensity score assigned, an important difference of -1 in the value 1 could be observed: the readers disagreed on whether the test line was present or not and, thus, if the test was positive or negative.

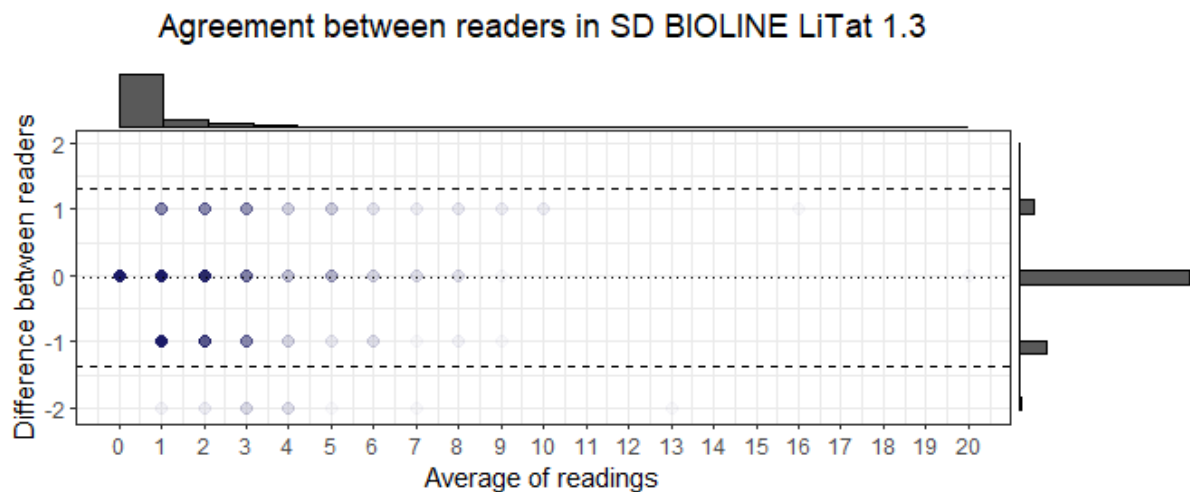


Figure 7. Agreement between readers in SD BIOLINE HAT test line 1, for the antigen LiTat 1.3. The dotted line represents the mean difference, while the dashed lines represent the mean plus or minus two times the standard deviation. The histogram on top represents how common each value was in the tests, and the histogram on the right how common the discordance between readers was.

For the second test line, containing the antigen LiTat 1.5 and represented in **Figure 8**, the average difference was 0.31, with a standard deviation of 1. A wider range of intensity scores was recorded for LiTat 1.5, and thus, the disagreement between readers was higher and more dispersed through the graph than for LiTat 1.3. The most frequent result was still zero, and just like it happened with LiTat 1.3, there was a frequent disagreement between the intensities scores of 1 and 0.

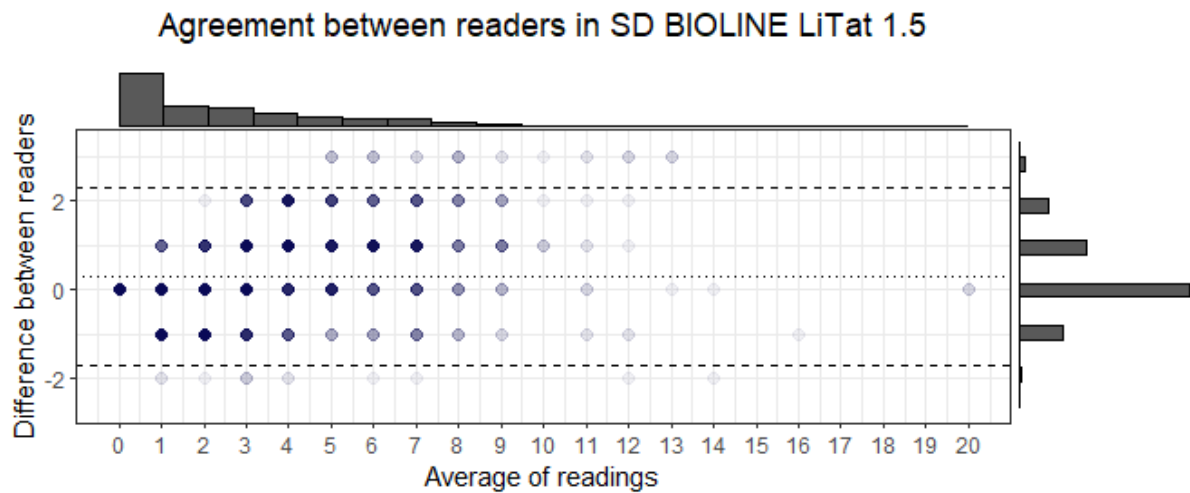


Figure 8. Agreement between readers in SD BIOLINE HAT test line 2, for the antigen LiTat 1.5. The dotted line represents the mean difference, while the dashed lines represent the mean plus or minus two times the standard deviation. The histogram on top represents how common each value was in the tests, and the histogram on the right how common the discordance between readers was.

The disagreement recorded in the intensity scores 0, 1 and even 2 in both test lines of the rapid test SD BIOLINE HAT was of special interest, since it could potentially lead to false negative results. To further evaluate the effect of these faint lines (lines of low intensity score) in the performance of the test, a new evaluation was carried out, where they were considered negative. As expected, a higher amount of negative results were reported, especially in later follow-ups (**Figure 9**): over 65% of the samples taken two years after treatment were negative if the intensity score of 1 was considered negative. For an intensity score of 2, the increase is even more pronounced: over three out of every four tests came back negative.

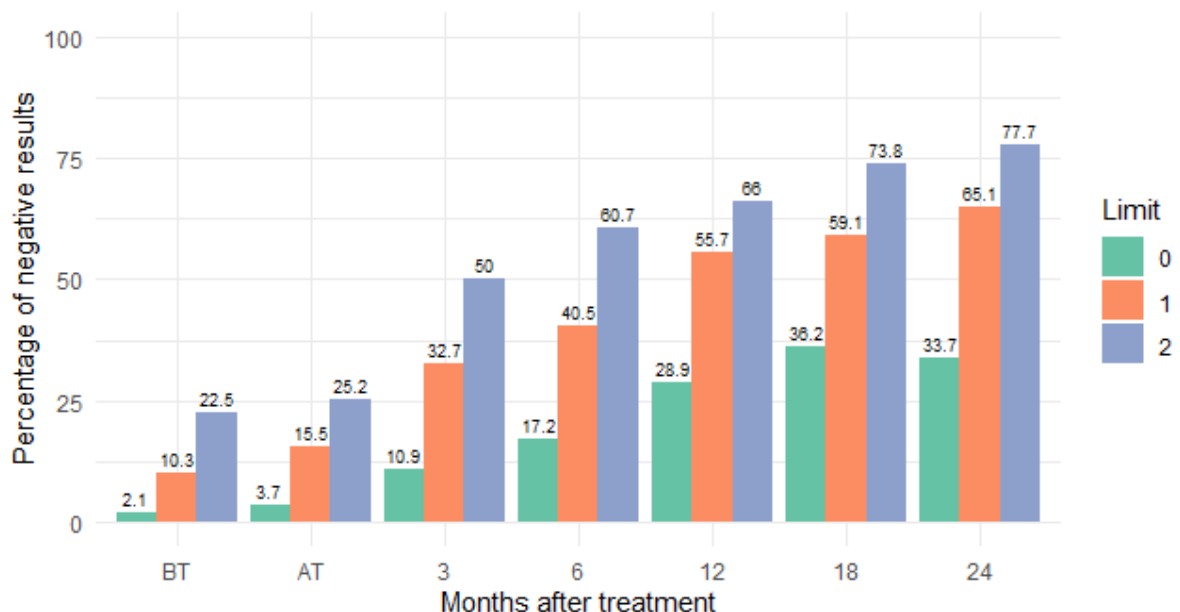


Figure 9. Percentage of negative tests in SD BIOLINE HAT per date, using three different thresholds to represent the three lowest intensities recorded (zero, one and two). BT = Before treatment, AT = After treatment, 3 = 3-months follow-up, 6 = 6-months follow-up, 12 = 12-months follow-up, 18 = 18-months follow-up, 24 = 24-month follow-up.

The impact of considering faint lines as negative results was further studied by analysing both test lines separately (**Table 3**) since, as mentioned before, low intensity scores were more common for the antigen LiTat 1.3 (test line 1). If faint lines were considered as negatives, the sensitivity of test line 1 in BT samples lowered dramatically: from detecting 7 out of every 10 positive results with the cut-off 0, to just 5 out 10 with the cut-off at 1. If lines with an intensity score of 2 were also counted as negative, the first test line only detected 3 out of 10 positive results. The effect of faint lines was especially important in samples taken later in time: for positive thresholds of 0, 1 and 2, LiTat 1.3 in SD BIOLINE HAT only detected 18, 4 and 1% of the results as positive, respectively.

Table 3. Number of negative results at each date of the follow-up in SD BIOLINE HAT, with different cut-offs (set at intensity scores 0, 1 and 2) and individually represented for each test line and the global score of the test.

		Before treatment (n=329)*	After treatment (n=321)	3 months (n=312)	6 months (n=266)	12 months (n=193)	18 months (n=150)	24 months (n=166)
Positive > 0	Test	7 (98%)	12 (96%)	34 (89%)	46 (83%)	56 (71%)	54 (64%)	56 (66%)
	LiTat 1.3	90 (72%)	98 (69%)	159 (49%)	177 (33%)	150 (22%)	118 (21%)	136 (18%)
	LiTat 1.5	7 (98%)	13 (96%)	35 (89%)	49 (82%)	57 (70%)	59 (61%)	60 (64%)
Positive > 1	Test	34 (90%)	50 (84%)	102 (67%)	108 (59%)	108 (44%)	88 (41%)	108 (35%)
	LiTat 1.3	164 (50%)	189 (41%)	251 (20%)	228 (14%)	178 (8%)	134 (11%)	159 (4%)
	LiTat 1.5	41 (87%)	58 (82%)	111 (64%)	116 (56%)	113 (41%)	94 (37%)	111 (33%)
Positive > 2	Test	74 (76%)	81 (75%)	156 (50%)	162 (39%)	128 (34%)	110 (27%)	129 (22%)
	LiTat 1.3	216 (34%)	242 (25%)	279 (11%)	245 (8%)	186 (4%)	145 (3%)	164 (1%)
	LiTat 1.5	85 (74%)	91 (72%)	169 (46%)	168 (37%)	130 (33%)	112 (25%)	131 (21%)

* n = number of patients in each follow-up event. In the table, number of negative results per test in n samples analysed, in parenthesis, percentage of positive samples in the same follow-up visit.

For LiTat 1.5, contained in the test line 2 of SD BIOLINE HAT, the effect of faint lines also had an impact on the outcome of the test, but it is not as acute as in LiTat 1.3. For the samples BT, the sensitivity of the second test line fell from 98% to 87% and 74%, with cut-offs at 1 and 2, respectively. In samples taken two years after treatment, the effect of fainter test lines was again more pronounced, which was normal considering that later follow-ups scored lower intensity values: the number of negative tests increased as the threshold of positivity raises. If the intensity score of 1 was considered negative, only 33% of the samples were positive, while for a threshold of 2 the positivity fell to 21%.

b. ELISA/T. *b. gambiense*

For ELISA/T. *b. gambiense*, variations from the original protocol related to the concentration of the positive and negative control led to most of the results, expressed as PP, having values well over 100%. The main consequence of this is that the former cut-off value for what was considered positive, set in 35, became impractical. New thresholds were

considered, based on the performance of the ELISA/T. *b. gambiense*, with the same concentrations of positive and negative controls, had in another collection of samples that combined known positives and negatives (see Appendix 2). The four thresholds considered were 80, 100, 120 and 150. Higher cut-off values meant higher specificities at the cost of lower sensitivities. Indeed, as shown in **Table 4**, higher cut-off values led to more negative results, especially in follow-ups later in time. Finally, after the analysis described in Appendix 2, the chosen threshold for the rest of the analysis was 120.

Table 4. Number of negative results detected by the test ELISA/T. *b. gambiense*, separated by follow-up date.

	Before treatment (n=329)*	After treatment (n=321)	3 months (n=312)	6 months (n=266)	12 months (n=193)	18 months (n=150)	24 months (n=166)
Threshold 80	10 (97%)	7 (98%)	14 (96%)	21 (92%)	24 (88%)	28 (81%)	35 (79%)
Threshold 100	17 (95%)	15 (95%)	23 (93%)	36 (86%)	38 (80%)	45 (70%)	57 (66%)
Threshold 120	24 (93%)	21 (93%)	46 (85%)	67 (75%)	64 (67%)	60 (60%)	78 (53%)
Threshold 150	44 (87%)	48 (85%)	86 (72%)	104 (61%)	94 (51%)	93 (38%)	108 (35%)

* n = number of patients in each follow-up event. In the table, number of negative results per test in n samples analysed, in parenthesis, percentage of positive samples in the same follow-up visit.

The distribution of the PP values per follow-up date, as seen in **Figure 10**, shed more light on the performance of this test. Once again, a decreasing tendency could be observed in the PP values as time passed: starting with a median of 231.9 (IQR: 173.5-278.1) in the BT samples, it went steadily and significantly down in all the follow-ups. For AT samples, the median was 218.8 (IQR: 170.4-266.5, $p<0.001$); 3 months after treatment, the median was 187 (IQR: 142.2-241.2, $p<0.001$); after 6 months it lowered to 171.6 (IQR: 120-224.6, $p<0.001$), and after one year to 150.7 (IQR: 107.1-202.4, $p<0.001$). The samples after 18 months (131.8, IQR: 89-181.9, $p<0.001$) and two years (124.1, IQR: 87.9-170.6, $p<0.001$) showed a further decrease.

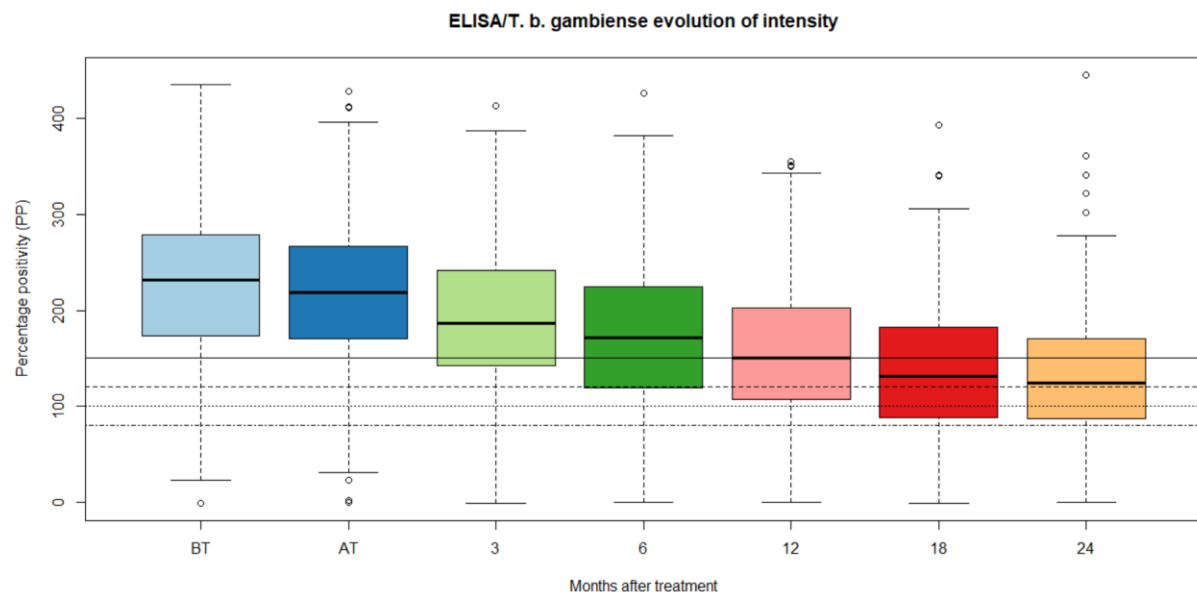


Figure 10. Evolution of antibody binding over time in ELISA/T. *b. gambiense*. The different thresholds considered are marked as horizontal lines in the graph: the dotted and dashed line sets the threshold at 80, the dotted line at 100, the dashed line at 120, and the solid line at 150. BT = Before treatment, AT = After treatment,

3 = 3-months follow-up, 6 = 6-months follow-up, 12 = 12-months follow-up, 18 = 18-months follow-up, 24 = 24-month follow-up.

c. INHIBITION ELISA

In the case of the newly developed inhibition ELISA, the number of negative results through all the follow-up was much lower than in the other techniques, with 92% of the samples still being detected as positive after two years, as seen in **Table 5**. Again, this test evaluates separately the antibody response against the antigens LiTat 1.3 and LiTat 1.5: although in some samples (namely, those taken before treatment, after treatment, 3 months and 24 months after treatment, with p-values of 0.44, 0.72 and 0.08) the performance of both antigens was equal, in the others LiTat 1.3 reported significantly less positive results than LiTat 1.5 (p-value = 0.01, 0.012 and 0.0455 for 6, 12 and 18 months after treatment, respectively).

Table 5. Negative results detected by Inhibition ELISA per follow-up date. Negative results detected by each antigen (LiTat 1.3 and LiTat 1.5) per follow-up date.

	Before treatment (n=329)*	After treatment (n=321)	3 months (n=312)	6 months (n=266)	12 months (n=193)	18 months (n=150)	24 months (n=166)
Inhibition ELISA	6 (98%)	1 (100%)	3 (99%)	4 (99%)	10 (95%)	7 (95%)	13 (92%)
LiTat 1.3	11 (97%)	6 (98%)	18 (94%)	29 (89%)	36 (81%)	25 (83%)	39 (77%)
LiTat 1.5	8 (98%)	4 (99%)	8 (97%)	13 (95%)	20 (90%)	14 (91%)	27 (84%)

n = number of patients in each follow-up event. In the table, number of negative results per test in n samples analysed, in parenthesis, percentage of positive samples in the same follow-up visit.

As far as how the percentage of inhibition evolved over time, both antigens performed the same, with a steady and significant decrease (all p-values<0.05) over time, but with even the lowest quartiles of the later follow-ups remained above the positive threshold. For LiTat 1.3, the median of BT samples is 79.7 (IQR:69.1-87.6), 77.5 (IQR:65.9-85.9) for AT, 69 (IQR:52-80.2) after three months and 64 (IQR:46.5-76.1) after six months. In the last three follow-ups, a year, a year and a half and two years later, the median percentage of inhibition values lowered to 57.2 (IQR:35.9-70.9), 54.5 (IQR:35.8-66.6) and 49.7 (IQR:30.9-64.5), respectively.

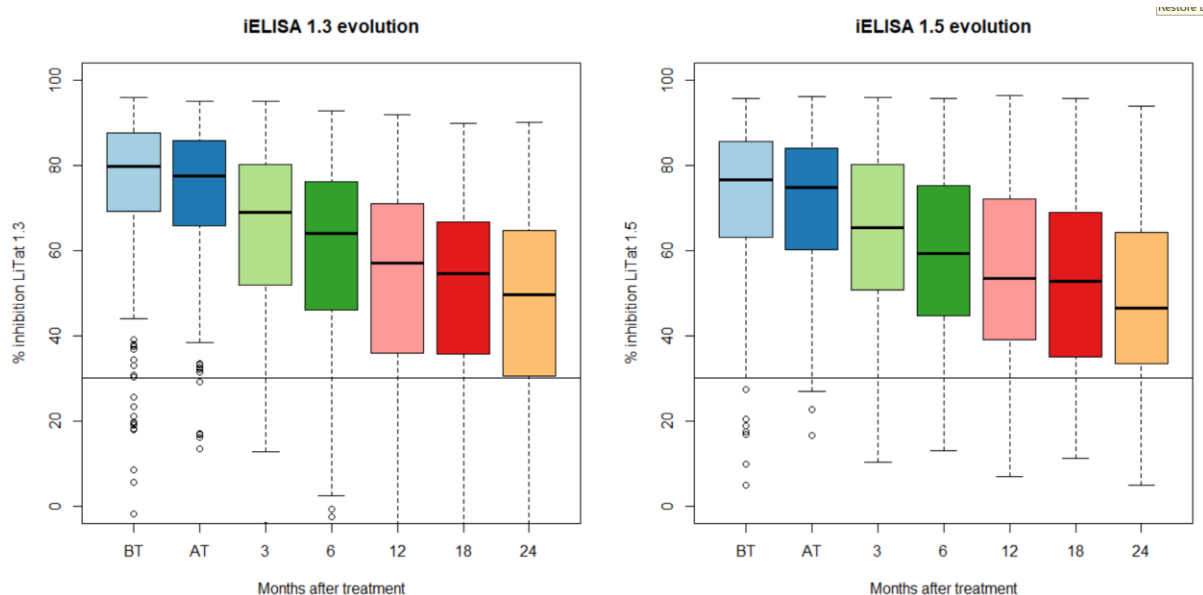


Figure 11. Evolution of the percentage of inhibition to antigens LiTat 1.3 (left) and LiTat 1.5 (right) registered during the follow-up. The black line marks the cut-off of what is considered positive and negative at 30% inhibition in both antigens BT = Before treatment, AT = After treatment, 3 = 3-months follow-up, 6 = 6-months follow-up, 12 = 12-months follow-up, 18 = 18-months follow-up, 24 = 24-month follow-up.

For the antigen LiTat 1.5, the samples BT and AT reported medians of 76.6 (IQR: 63.1-85.6) and 74.9 (IQR: 60.1-84) each. For the rest of the follow-ups, the medians were significantly lower, compared to that of the BT samples ($p < 0.001$): at 3 months, the median was 65.5 (IQR: 50.8-80.1); at 6 months, 59.2 (IQR: 44.8-75.4); at 12 months, 53.5 (IQR: 39.1-72.2); at 18 months, 52.7 (IQR: 35.1-68.8); and at 24 months, the median was 46.6 (IQR: 33.5-64.1). In fact, the trends of the percentage of inhibition for both antigens were so similar they were comparable: although the values of the samples BT and AT were significantly different ($p\text{-value} < 0.001$ and $p\text{-value} = 0.01$ each) in LiTat 1.3 and LiTat 1.5, there was no significant difference in the rest of the follow-ups.

d. IMMUNE TRYPANOLYSIS TEST

As mentioned before, the immune trypanolysis test was only performed for variant LiTat 1.5 in all the samples of this study, since LiTat 1.3 had previously been performed [Ref]. However, LiTat 1.3 had only been performed in the samples of cured patients (163 of the total), so for those patients that relapsed, died or stopped showing up to the follow-up, only the results of the antigen LiTat 1.5 were available. Due to the risk of performing this test and the laboratory animals involved, it was finally decided not to perform LiTat 1.3 TL in all the missing samples; instead, it was performed only in the BT samples. Thus, those samples negative for LiTat 1.5 that have no result for LiTat 1.3 cannot be considered as negative. To adapt to this limitation, separate analysis were performed for cured patients (with data for both antigens) and for those with incomplete follow-up (only results for LiTat 1.5).

Table 6. Number of positive and negative results detected by the immune trypanolysis test globally and separated in the antigens LiTat 1.3 and LiTat 1.5 in **cured patients**.

Before treatment (n=329)*	After treatment (n=159)	3 months (n=156)	6 months (n=156)	12 months (n=145)	18 months (n=132)	24 months (n=159)
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TL	329/0 (100%)	159/0 (100%)	155/1 (99%)	148/3 (98%)	134/11 (92%)	120/12 (91%)	136/23 (86%)
LiTat 1.3	326/3 (99%)	158/1 (99%)	152/4 (97%)	148/8 (95%)	128/17 (88%)	113/19 (86%)	126/33 (79%)
LiTat 1.5	304/25 (92%)	144/15 (91%)	129/27 (83%)	115/41 (74%)	90/55 (62%)	84/48 (64%)	95/64 (60%)

* n = number of patients in each follow-up event. In the table, number of positive/negative results per test in n samples analysed, in parenthesis, percentage of positive samples in the same follow-up visit.

The results of the samples where trypanolysis was fully performed (BT samples for all patients, follow-ups for cured patients) were summarized in **Table 6**. All the samples before treatment were positive for at least one antigen (3 of them were negative for LiTat 1.3 and 25 for LiTat 1.5). For the follow-ups, the percentage of positive samples decreased as time passed, with 86% of the samples remaining positive 24 months after finishing the treatment. The separate evaluation of LiTat 1.3 and LiTat 1.5 showed a significantly different performance ($p < 0.001$) between antigens. The decrease in positivity was steeper for LiTat 1.5: only 60% of the patients were positive after 24 months, compared to the almost 80% that remained positive in the same date for LiTat 1.3.

As for the evolution for intensity, the antigen LiTat 1.3 followed a particular trend: the median for all follow-ups was 100 (IQR: 100-100, except for the samples 24 months after the treatment, for which IQR: 90-100), as seen in **Figure 12**. For the antigen LiTat 1.5, the trend was more or less the same, but with a larger interquartile variation: the median for all follow-up dates was 100 (BT, AT IQR: 100-100; 3 months IQR: 80-100, 6 months IQR: 50-100, 12 months IQR: 10-100; 18 months, IQR: 0-100), except for 24 months after treatment, where the median was 90 (IQR: 0-100). For LiTat 1.3, values under 100 were rare enough to be considered outliers. In LiTat 1.5, the same was true for the first two samples (BT and AT), but after three months, the variability of the samples and the frequency of lower values increased enough to expand the first quartile of all the groups.

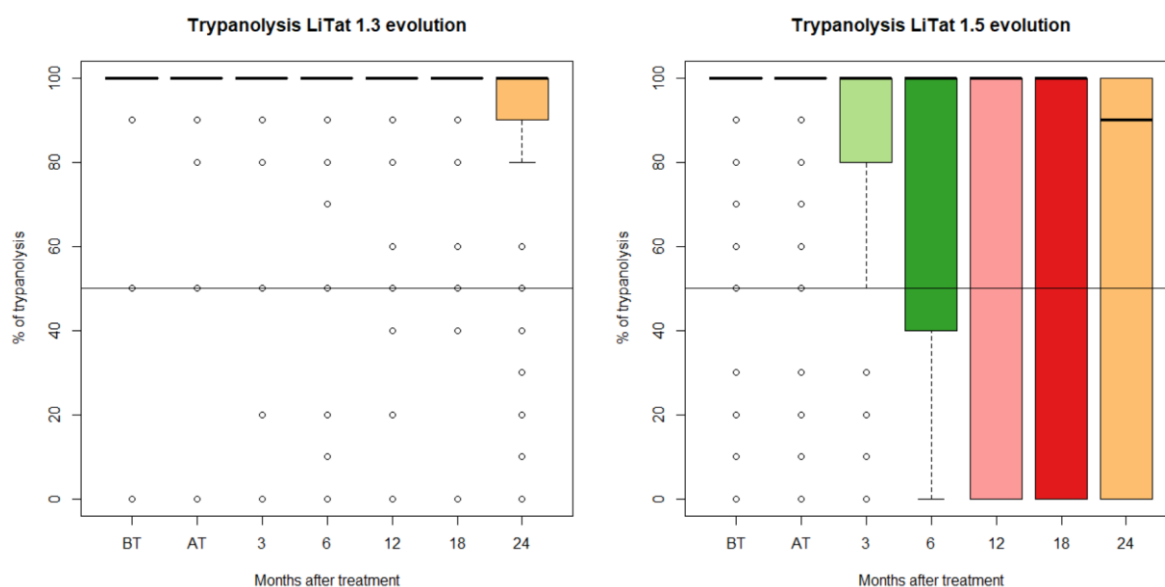


Figure 12. Evolution of the percentage of trypanolysis over follow-up date. On the left, results of antigen LiTat 1.3 in cured patients, on the right, LiTat 1.5 in cured patients. BT = Before treatment, AT = After treatment, 3 =

3-months follow-up, 6 = 6-months follow-up, 12 = 12-months follow-up, 18 = 18-months follow-up, 24 = 24-month follow-up.

There were a remaining 166 patients that relapsed, died or were lost during follow-up, whose samples were tested for the antigen LiTat 1.5, but had no data available for LiTat 1.3. The results are summarized in **Table 7**: the analysis of how the number of negatives evolve through time in this group cannot be carried out, since most of the patients with treatment failure relapse within a year, and the number of samples in later follow-ups was not big enough to be representative. The relapses explained the higher percentage of positive in this group of patients 3 and 6 months after treatment, compared to that of cured patients (**Table 6**).

Table 7. Results of trypanolysis test LiTat 1.5 in patients that were not cured.

	Before treatment (n=166)*	After treatment (n=162)	3 months (n=156)	6 months (n=110)	12 months (n=48)	18 months (n=18)	24 months (n=7)
LiTat 1.5	158/8 (95%)	153/9 (94%)	129/27 (83%)	87/23 (79%)	41/7 (85%)	12/6 (66%)	4/3 (57%)

* n = number of patients in each follow-up event. In the table, number of positive/negative results per test in n samples analysed, in parenthesis, percentage of positive samples in the same follow-up visit.

e. EVOLUTION OF THE ANTIBODY RESPONSE

Next, all the tests were compared with one another, in order to analyse how the seropositivity of cured patients is recorded in each method. The immune trypanolysis test was used as reference, since it is considered the most accurate test available today. Using the Kaplan Meier survival function, the probability of each test of remaining positive over time was plotted in **Figure 13**. Trypanolysis, the reference test, stayed positive almost 8 out of every 10 times (0.794, 95% CI = 0.74, 0.852) after two years, almost like inhibition ELISA (0.871, 95% CI = 0.828, 0.916). For the rest of the tests, HAT Sero-K-SeT had higher probabilities of remaining positive than trypanolysis (0.951, 95% CI = 0.92, 0.982), whereas ELISA/*T. b. gambiense* and SD BIOLINE HAT had higher probabilities of becoming negative: ELISA had a probability of remaining positive of 0.323 (95% CI = 0.274, 0.38), and SD BIOLINE HAT of 0.444 (95% CI = 0.39, 0.506). This RDT was the only test more likely to become negative than to stay positive two years after treatment.

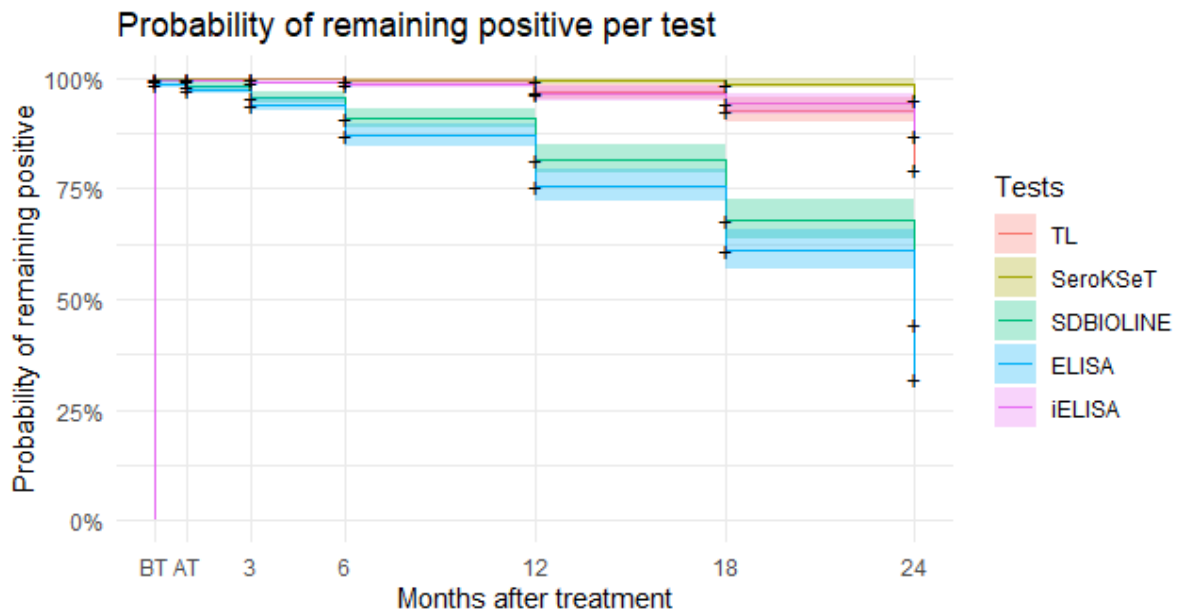


Figure 13. Probability of trypanolysis test, HAT Sero-K-SeT, SD BIOLINE HAT, ELISA/*T. b. gambiense* and inhibition ELISA of remaining positive through time.

Since SD BIOLINE HAT, inhibition ELISA and trypanolysis test the antigens separately before reaching a global result, it is also possible to study how the antibody responses for each antigen evolve through time in cured patients, and the probability of each antigen remaining positive through time. Again, the results of trypanolysis will be used as the standard against which the other two are compared.

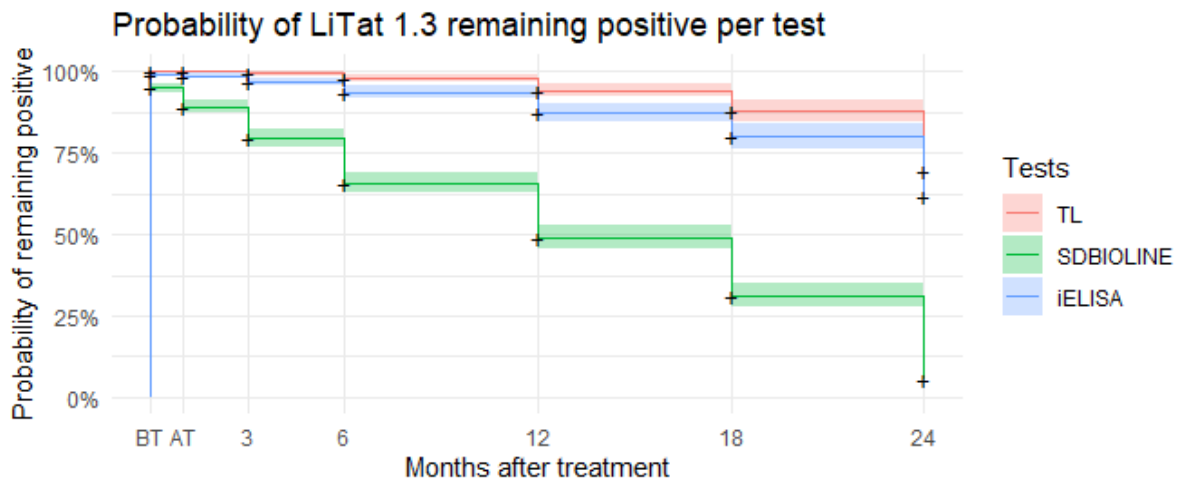


Figure 14. Probability of trypanolysis test, SD BIOLINE HAT and inhibition ELISA of remaining positive for antibodies against LiTat 1.3 through time.

As shown in **Figure 14**, SD BIOLINE HAT practically always became negative for antigen LiTat 1.3 after two years: the probability of it remaining positive was barely 0.055 (95% CI = 0.038, 0.078). iELISA had a probability of staying positive of 0.614 (95% CI = 0.557, 0.677), whereas trypanolysis had a probability of 0.697 (95% CI = 0.638, 0.761). In other words, there was a very small difference in the probabilities of remaining positive between inhibition ELISA

and the reference test, with sometimes overlapping confidence intervals, whereas SD BIOLINE HAT followed a completely different pattern.

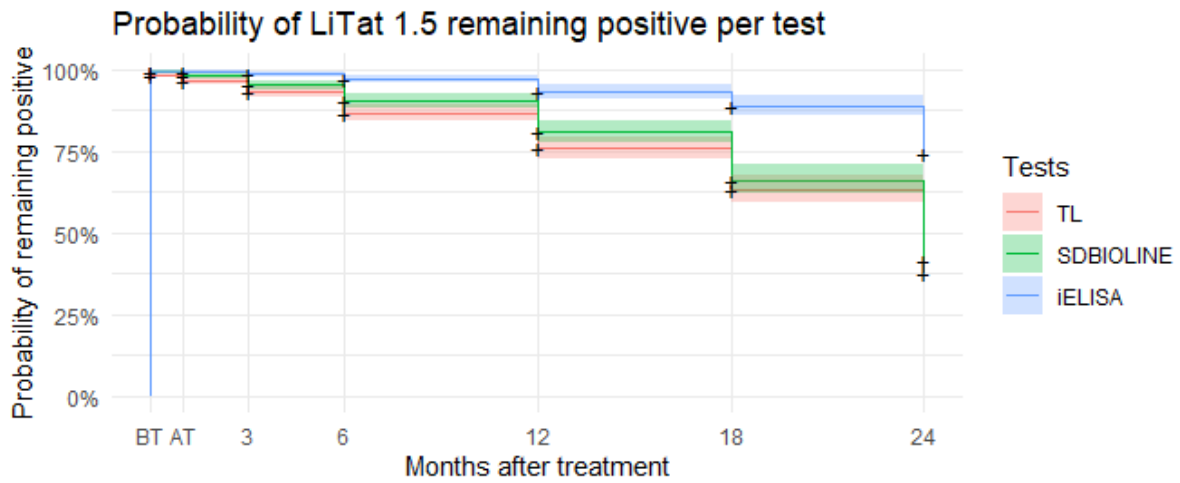


Figure 15. Probability of trypanolysis test, SD BIOLINE HAT and inhibition ELISA of remaining positive for antibodies against LiTat 1.5 through time.

The probability of becoming negative for the antigen LiTat 1.5 followed a completely different trend: in this case, inhibition ELISA had a probability of remaining positive of 0.744 (95% CI = 0.689, 0.803), higher than it had for the antigen LiTat 1.3 (0.614). However, SD BIOLINE HAT is more likely to remain positive for antigen LiTat 1.5 than it was for LiTat 1.3 (0.417, 95% CI = 0.364, 0.479, vs. 0.055), although still over half of the tests will become negative. Lastly, the reference test, trypanolysis, is much more likely to become negative for LiTat 1.5 than for LiTat 1.3, with a probability of 0.379 (95% CI = 0.327, 0.437), rather than the 0.697 shown before. Thus, for LiTat 1.5, it was SD BIOLINE HAT and trypanolysis that had overlapping probabilities and confidence intervals of remaining positive, whereas inhibition ELISA is much less probable of becoming negative than the other two.

VII. DISCUSSION

Gambiense human African trypanosomiasis is a fading disease, whose patient numbers do nothing but decrease year after year, thanks to the important investment and effort deployed by international non-profit organizations, bilateral country cooperation, health care services of endemic countries and the World Health Organization [4]. However, this new epidemiological situation, characterized by very low disease incidence, creates new needs in terms of diagnosis and treatment: old diagnostic tools are being improved (like the use buffy coat in mAECT), and new ones are being developed (like RDTs or inhibition ELISA), with improved sensitivity and specificity to detect the few cases that remain [63]. However, the diagnostic algorithm has not changed: as explained in **Figure 2**, the first step is still the serological identification of HAT suspects, followed by the parasitological confirmation of infection. This process is slow and requires electricity, equipment and skills that are sometimes difficult to bring to endemic areas [64]. Traditional active screening with mobile teams is becoming less cost-efficient, and passive screening methods, based on the integration of new serological tests like RDTs in local health centres, are now recommended by the WHO as the preferred surveillance strategy [65]. In this new context, it is important to (re)assess the screening potential of the newly-developed

serological tools, and to define what strategy should be followed when a positive serological result is reported.

This study aimed to clear out how individuals that had previously suffered of g-HAT would react to new serological screenings, either passive or active. Is a positive serological result enough to treat a patient? What if that person had previously suffered of g-HAT infection before, but had been successfully treated? Do the available serological tests still detect the antibody response, even years after finishing the treatment? To this end, all five available serological tests (HAT Sero-K-SeT, SD BIOLINE HAT, ELISA/*T. b. gambiense*, inhibition ELISA and immune trypanolysis) were tested in the THARSAT collection, that dates back to 2005 and followed a cohort of over 300 patients for two years after treatment. The main findings are i) that HAT Sero-K-Set and SD BIOLINE HAT have different performances, especially in samples collected two years after treatment, and ii) that inhibition ELISA's performance is comparable to that of trypanolysis.

In the first place, HAT Sero-K-SeT reported positive results for the vast majority of the samples tested. In fact, this rapid test reported no negative results in the samples before or just after treatment, meaning it successfully identified as positive all the samples of the patients before they were treated. The only 11 cases reported as negative belonged to the follow-up period, and six of them were to samples taken two years after treatment. Still, 96% of the samples remained positive two years after the treatment ended. On the other hand, SD BIOLINE HAT's negative results were much more common, especially, but not limited, during the later follow-up period (12, 18 and 24 months after the treatment). It reported 7 negative results in the samples before treatment, and two years after treatment, 34% of the samples had become negative in SD BIOLINE HAT. Besides, the characteristic that tells SD BIOLINE HAT apart from the other rapid test is that it has separate lines for antigens LiTat 1.3 and LiTat 1.5. These two test lines have different performances: LiTat 1.3 has a higher negative rate than LiTat 1.5; only 18% of the samples were positive for LiTat 1.3 after two years, compared to the 64% that were positive for LiTat 1.5 in the same follow-up timepoint.

Since both RDTs are immunochromatographic tests, besides the presence or absence of the test lines, it is also important to consider their intensity. HAT Sero-K-SeT reported high intensity scores; although they faded over time, most intensities were well over 7 (colour scale provided in **Figure 3.A**), even in later follow-ups. This means that the test lines were clearly visible even when the samples had been collected two years after the infection. Besides, high agreement between readers was also achieved in this test (disagreement of $-0.41 (\pm 0.95)$), so the readings of HAT Sero-K-SeT were both clear and precise. SD BIOLINE HAT, however, reported much lower intensities in both test lines: although the test line for LiTat 1.5 was more intense than that of LiTat 1.3, it was still low (around 5, compared to the median intensity of 1 recorded for LiTat 1.3) and faded over time. While the agreement between readers was again high for both test lines, there is an important source of disagreement in the values 0, 1 and 2: the lines are so faint in this range of the colour scale (provided in **Figure 3.B**) that disagreement between readers is more common. The problem is that disagreement in these values can give place to false negative results and, thus, misdiagnosis.

Indeed, if these faint lines are also considered as negative results, the number of negatives reported by SD BIOLINE HAT increases, especially later during follow-ups: instead of 33.7% of negative results, it goes up to 65.1 or even 77.7% of negative results, depending on where the cut-off is placed. Again, there is a difference in the test lines for LiTat 1.3 and LiTat 1.5: as mentioned before, LiTat 1.3 reported very low intensity scores, so if faint lines are also considered negative, only 1 to 4% of the tests will be positive for this line. For LiTat 1.5, the effect of faint lines is less pronounced but acute nonetheless: between 70 and 80% of the samples two years after treatment will be negative. Even with stricter criteria, the outcome of LiTat 1.5 has the same or better performance than LiTat 1.3. In fact, LiTat 1.5 positivity through the different cut-offs and follow-up dates follows the same patterns as the global test result, meaning that the ultimate result of the RDT heavily depends on the outcome of the second test line, and LiTat 1.3 has very little influence.

Just as the RDTs, ELISA/*T. b. gambiense* also shows lower values of percentage binding in samples belonging to later dates in the follow-up (as seen in **Figure 10**). The number of negative results registered, however, depends on where the threshold is placed: to this end, a separate analysis was carried out to find the threshold with the highest sensitivity and specificity (**Appendix 2**). However, this analysis was carried out in a completely different collection of samples. When brought to the THARSAT collection, almost 50% of the samples of the samples collected two years after the patients had finished treatment became negative, although it is true that it had a good performance in the samples before treatment, detecting 93% as positive. Moving the threshold up and down the Y axis changes the proportion of positive and negative results found (**Table 4**), and opens the door to adapting this test to different epidemiological situations; a lower threshold (around 80) will be able to detect the highest proportion of seropositive individuals, at the cost of misdiagnosing as positive seronegative individuals, while a stricter threshold (around 150) can be useful for the detection of true positives, although it will lose a few of them. However, further trials with a standardized protocol are necessary before even thinking of applying this technique in the field.

Inhibition ELISA, despite also showing lower percentage of inhibition in later follow-ups, had a completely different outcome compared to ELISA/*T. b. gambiense*. Not only did it detect 98% of the samples before treatment as positive, but it also remained positive in 92% of the samples taken two years after finishing the treatment. iELISA also works separately with both antigens; although LiTat 1.3 and LiTat 1.5 have significantly different performances in 4 out of 7 timepoints, the differences are much less accentuated than for SD BIOLINE HAT. For example, in the samples collected 18 and 24 months after the infection, LiTat 1.3 detected 83 and 77% of the samples positive, while LiTat 1.5 detected 91 and 84% as positive. However, the global outcome for the test in those timepoints is 95 and 92% of positivity; neither antigen has a better performance than the other, rather they complement each other and offer a better result when analysed together than alone. The negative percentage of inhibition values observed for LiTat 1.3 in later follow-ups (**Figure 11**) are considered a normal phenomenon for this antigen, as reported before by Geerts et al. in [34], and do not influence in the final outcome of the test.

Finally, the results of trypanolysis are tricky to compare, since the test was only fully performed in about half of the patients. In these, trypanolysis was 100% positive for all the samples before treatment, and remained positive in 86% of the samples taken two years afterwards. LiTat 1.3 and LiTat 1.5 had significantly different results again, but this time LiTat 1.3 had a higher positivity rate than LiTat 1.5: both antigens had high positivity rates in the samples before and after treatment, that lowered as the follow-up progressed. Again, neither antigen perfectly mimicked the global result, so it is possible that some samples had antibodies with better affinity for LiTat 1.3 than 1.5, and vice versa. The differences between the antigen performance between inhibition ELISA and trypanolysis can be explained by set-up of each test: the immunoassay works with isolated VSGs [Supplementary material 3 of Geerts et al.], while trypanolysis makes use of cloned trypanosomes expressing each antigen. Although they are technically the same proteins, different epitopes are expected to be exposed in each test, leading to differences in the binding of antibodies.

On the other hand, the samples that were only analysed for LiTat 1.5 trypanolysis showed similar percentages of positivity compared to those LiTat 1.5 results of cured patients. However, in the absence of LiTat 1.3 information, the final diagnosis cannot be performed. Since among these samples were those belonging to patients who had relapsed, the differences between the evolution of seropositivity in cured and relapse patients could not be assessed, in the absence of results from the golden standard test. However, previous studies already point to a lack of correlation between the serological status and treatment outcome [66], [67], although it is also true these studies have only been performed using CATT.

When comparing all tests together, based on the probability of remaining positive two years after the treatment (**Figure 13**), inhibition ELISA and trypanolysis are the only ones with overlapping trends, meaning both tests are equally likely to remain positive even years after the infection was solved. These results validate iELISA as a much more accessible and easier to perform alternative to trypanolysis. HAT Sero-K-SeT, on the other hand, is the test most likely to remain positive in the long term (0.951, 95% CI = 0.92, 0.982), more so than trypanolysis. SD BIOLINE HAT and ELISA/*T. b. gambiense*, on the other hand, are very likely to become negative after a time: the probabilities of staying positive after two years are 0.444 and 0.323 respectively. The probability of staying positive when studying the antigens separately in SD BIOLINE HAT, inhibition ELISA and trypanolysis are more reflection of the differences in the setup of each test, and confirm that the use of both LiTat 1.3 and LiTat 1.5 has a better performance than the use of either antigen alone.

To the best of our knowledge, this study is the first one to evaluate the performance of the five available serological alternatives to CATT (namely, HAT Sero-K-SeT, SD BIOLINE HAT, ELISA/*T. b. gambiense*, inhibition ELISA and immune trypanolysis test) in a collection of samples of treated patients through their follow-up. This study confirms previous evidence of long-lasting antibody response in treated g-HAT patients [14], [66], [68], [69], although they mostly used CATT and had generally smaller sample sizes. It also compares how different serological tests, using the same antigens, report variable results for the same collection of samples. However, it turned out that two years is too-short of a follow-up to verify if the samples become negative, since most tests still reported most samples as positive. Although

the intensity signal of each test has been compared over time (were it the intensity of the colour, the percentage of binding, of inhibition, or of trypanolysis), none of these tests are quantitative, so the changes in the intensity recorded for every test cannot be linked to a reduction of the antibody titre alone, since other factors, like presentation of critical epitopes in the test, stability of the antigen, reactivity and affinity of the antigen with specific antibodies, may have an effect [19]. Another limitation is that there is no way of telling if the samples that do become negative are truly negative, or if the antibody titres lower below the threshold of even the most sensitive test. Although it is true that trypanolysis is considered the golden standard, and that in theory its sensitivity and specificity are 100%, in practice it does have limitations.

The absence of negative controls in the tests performed makes it difficult to compare to other studies already published, since most of them calculate the sensitivity, specificity, positive and negative predictive value of each test. An approximate sensitivity calculation can be made, based on the results of each test on the samples before treatment, since they have all been confirmed as positive with parasitological tests. Thus, the sensitivity of HAT Sero-K-SeT would be 100% (compared to 98.5-99.1% published in [25], [70]); for SD BIOLINE, 98% (compared to 89.2, 92 or 99.6% published in [28], [30], [70], that also reflected differences in intensities between the test lines of LiTat 1.3 and LiTat 1.5); for ELISA/*T. b. gambiense*, 96% (compared to 98.7% published in [60]); for inhibition ELISA, 98% (the same as published in [34]); and for LiTat 1.3/LiTat 1.5 trypanolysis, 100% (compared to the 98.5% estimated in [31]).

This study aimed to shed some light to unanswered questions regarding g-HAT seropositivity in low-prevalence settings. Gambiense sleeping sickness has been endemic in the same areas for decades [10], so when performing a massive screening, it would be expected to find seropositive individuals that remained so after treatment: the results of this work confirm that treated individuals will remain seropositive at least two years after finishing the treatment, while other reports indicate the seropositivity could last even longer. In the context of incomplete or non-existent clinical records and a fading awareness of the disease, the identification of already-treated patients is important for the long-term surveillance programs. These patients will be positive in serological tests but negative in the confirmatory, parasitological tests. The current diagnostic algorithm says they should be tested again after six months, which is a waste of resources, since reinfection is extremely unlikely [71] and new treatment options show very low relapse rates [43], [46]. The preventive treatment of serologically-positive patients with negative parasitological tests is also heavily discouraged, due to the adverse effects even the safest option treatment presents [72], and due to the risk of drug resistance massive, indiscriminated treatment always entails.

In this context, this study shows that the two rapid diagnostic tests studied here, that are being considered for their implementation in first-line healthcare centres, have different responses to long-ago-treated patients: while HAT Sero-K-SeT will detect treated individuals as clearly positive, we would expect negative or very faint results in SD BIOLINE HAT. Hence, the combination of both RDTs could help clear out if a patient is a recent infection (both tests would be positive) or a former case (SD BIOLINE HAT would be negative or show very faint lines, while HAT Sero-K-SeT would remain positive). The idea of combining two

RDTs was already suggested in control programs for HIV [73], and this study is not alone in proposing a similar approach for gambiense HAT: in [70], Jammoneau et al. already suggested that the simultaneous use of both HAT Sero-K-SeT and SD BIOLINE HAT would improve the imperfect specificity of both RDTs. Although the cost-effectiveness of this strategy is yet to be evaluated, it would be a way to avoid over-treatment of the endemic population, which is also an expense, as well as to save both money and time in parasitological tests. Since budget constraints are always a factor, should we choose just one of the RDTs as the default screening method, HAT Sero-K-SeT would be recommended, seeing that it showed a better performance than SD BIOLINE HAT, with a much clearer outcome and a higher sensitivity.

Furthermore, this study also suggests that inhibition ELISA could fulfil the role that trypanolysis has now, confirming the presence of antibodies against *T. b. gambiense* in cases where the diagnosis is unclear [74]. Although the immune trypanolysis test remains the golden standard serological test, it is a highly complicated test that is ~~not expected to be performed but in only~~ three or four laboratories in the whole world. Inhibition ELISAs, on the other hand, could be expected to be implemented in peripheral health centres, since it does not require of ~~complicated~~ the technical ~~and human needs skills~~ that trypanolysis does, but it can provide almost identical sensitivity and specificity, saving a lot of time in transporting the samples. Gambiense inhibition ELISA has thus the potential to become a key tool in future surveillance programs for the interruption of transmission of gambiense sleeping sickness, combined with sensitive molecular techniques for the confirming diagnosis in peripheral health centres, while RDTs take the brunt of the screening in sentinel, first-line centres [75].

Although promising, future research regarding serological tests is still needed, ~~on-in~~ especially regards ~~to of~~ how the available serological tests will react to relapses. Though that data was briefly presented in **Table 7**, with relapse samples analysed in LiTat 1.5 trypanolysis, due to lack of time it was not further explored in this study. A more in-depth analysis of the results, both in the other tests and in trypanolysis, is needed to know what results the different serological tests are expected to report in the cases of treatment failure. It is true, however, that former studies show no correlation between serology and treatment outcome [66], and that the drugs that are currently been used (fexinidazole, pentamidine and NECT) have a much lower treatment failure than their predecessor, melarsoprol [45].

In the context of achieving interruption of transmission, the role of asymptomatic individuals and of infected animals in the transmission of *Trypanosoma brucei gambiense* is yet to be determined. Nonetheless, despite the questions that remained unanswered, huge steps have been taken recently towards the 2030 goal: the development of new diagnostic tests, both serological and molecular, can greatly help finding the few remaining cases of g-HAT, and new drugs, like fexinidazole or acoziborole, are expected to make treatment of both stages of the disease more affordable and less painful. With its incidence at an all-time low, the elimination of gambiense sleeping sickness can be very close, and research and disease surveillance are our best tools to get there.

VIII. REFERENCES

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APPENDIX 1 :
Colour scales provided by the manufacturers of
HAT Sero-K-SeT and SD BIOLINE HAT.

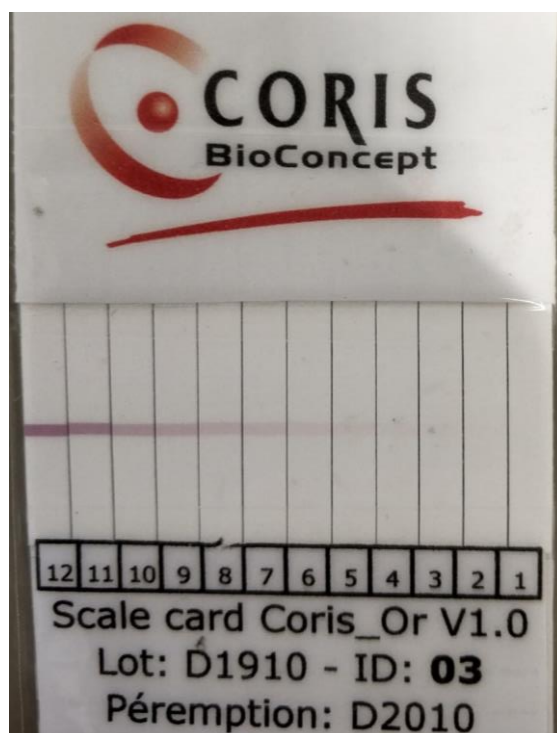


Figure 16. Colour scale for the rapid test HAT Sero-K-SeT, provided by the manufacturer CORIS BioConcept.

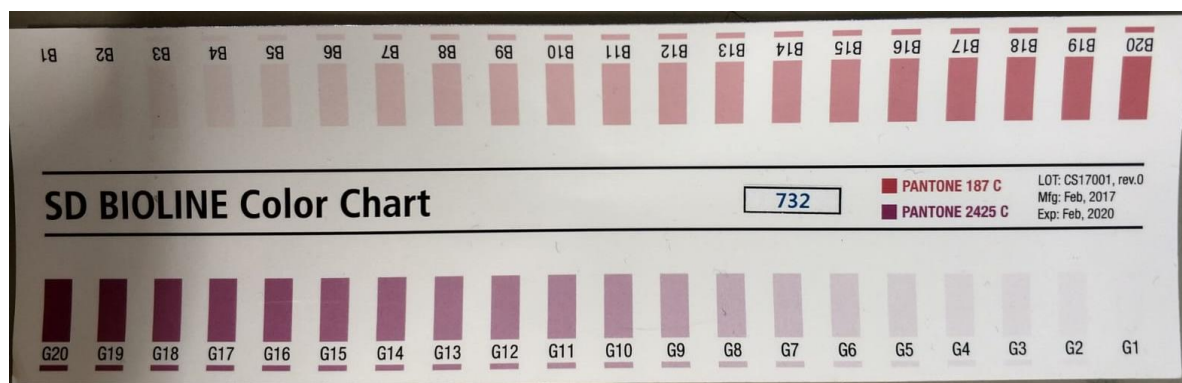


Figure 17. Colour scale for the rapid test SD BIOLINE HAT, provided by the manufacturer Standard Diagnostics.

APPENDIX 2 : **Determination of cut-off values for ELISA/*T. b. gambiense*.**

Due to a mistake in the concentrations of the positive and negative controls in the ELISA plates, the optical density (OD) values registered for the samples were higher than those registered for the positive control. Thus, the cut-off value marked by the protocol became obsolete, since almost all samples were over the threshold. In order to determine a new cut-off value, the same ELISA/*T. b. gambiense* test, using exactly the same protocol, was performed in another collection of samples. This new collection is made out of samples from g-HAT patients and controls from DRC, Guinea, Chad and Uganda, and belongs to the WHO Human African Trypanosomiasis Specimen Bank [76].

300 samples from this collection, distributed evenly as g-HAT cases and negative controls, were tested following exactly the same protocol as described in the Methods section. The results are summarized in Figure x.

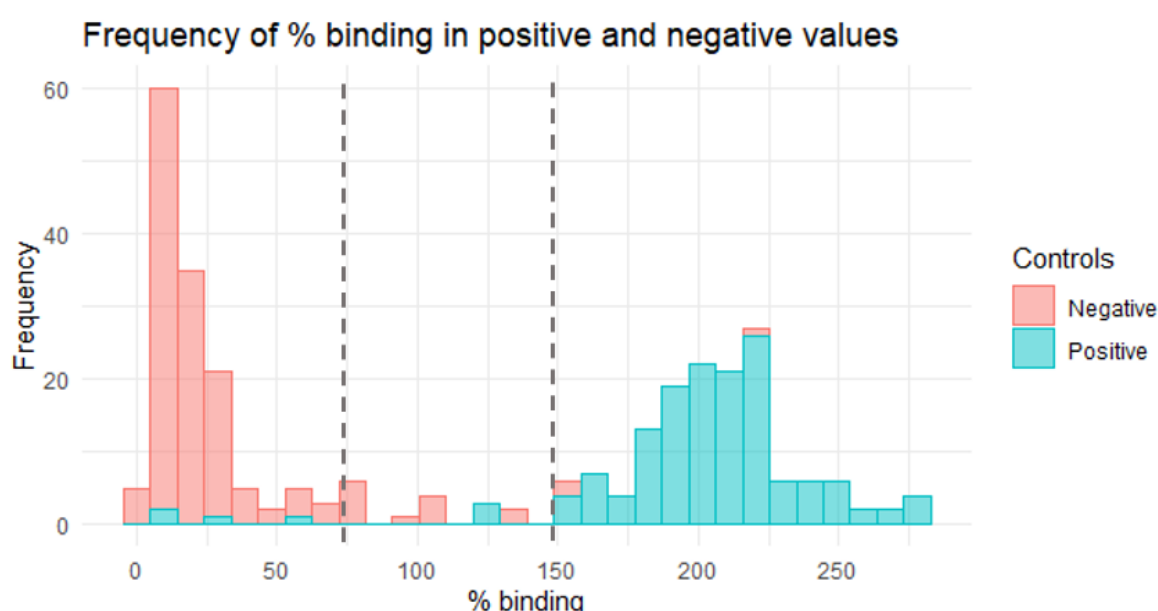


Figure 18. Distribution of percentage of binding in negative (red) and positive (blue) controls. Dashed lines mark the cut-off value range.

Using this same data with the “cutpointnr” package on RStudio Version 1.4.1106 (Ed. R Foundation for Statistical Computing, Vienna, Austria), the Youden index [77] was calculated, to find the cut-off value with the highest specificity and sensitivity. The result was 122.46, that has a sensitivity of 0.9732 and a specificity of 0.9662. Thus, the cut-off value used for this study was 120.